



WELCOME TO THE PYTHON DECAL!

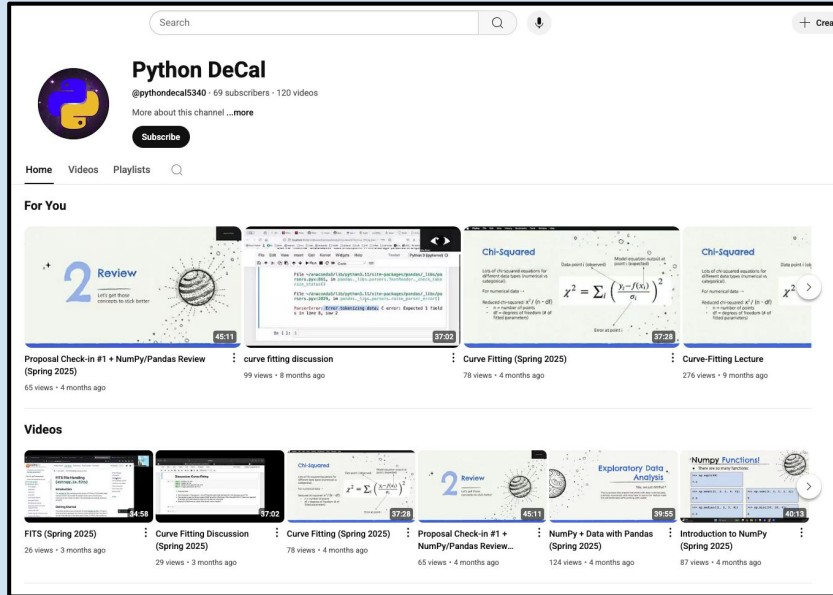
Lecture: GitHub
[Class 3] September 8th, 2025



LECTURES WILL BE RECORDED

This class will be
recorded and posted to
our YouTube Channel!

Subscribe :)



CLASS PLAYLIST ON SPOTIFY!



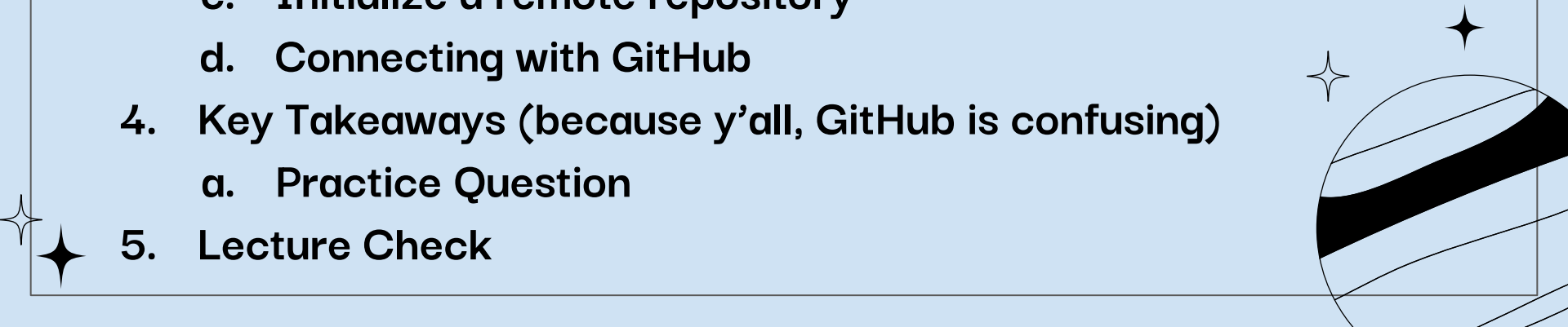
Scan the QR if you would like to see our class playlist!

We will try and play it before each class, remind us if we forget...

Also posted on bCourses



overview for today!

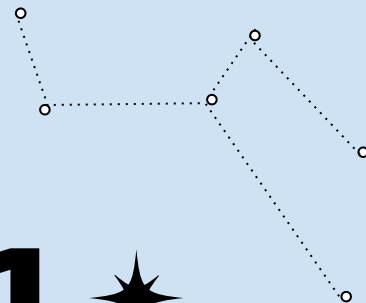
1. Announcements
 2. Warm-Up
 3. Demo
 - a. Initialize a local repository
 - b. Saving Your Work
 - c. Initialize a remote repository
 - d. Connecting with GitHub
 4. Key Takeaways (because y'all, GitHub is confusing)
 - a. Practice Question
 5. Lecture Check
- 

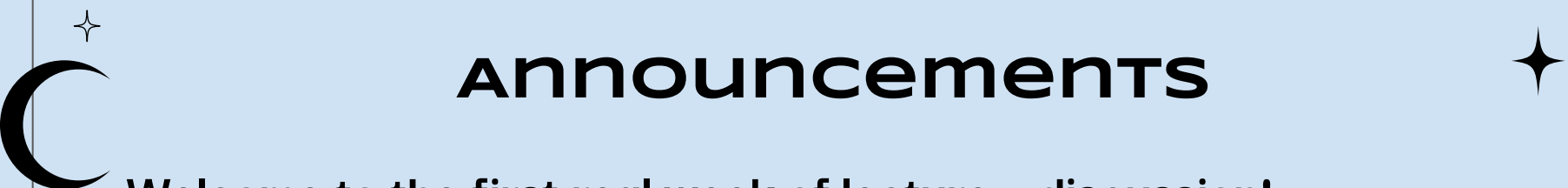


★ O1 ★

Announcements

:)) :)) :)) :)) :)) :)) :)) :)) :)) :))





Announcements

Welcome to the first real week of lecture + discussion!

This will be the typical format as we move throughout the semester.

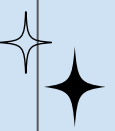
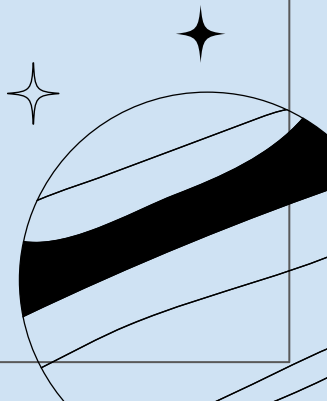
This week we will be focusing on GitHub.





Announcements



- HW #1 is due this Wednesday, Feb. 4th at 11:59 PM
 - HW #1 late due date is Feb. 18th at 11:59 PM
 - Next, next Wednesday
 - HW #2 was released today Monday, Feb. 2nd
 - HW #2 is due next Wednesday, Feb. 11th at 11:59 PM
 - HW #2 late due date is Feb. 25th at 11:59 PM
 - Office hours are being finalized this week!
- 
- 

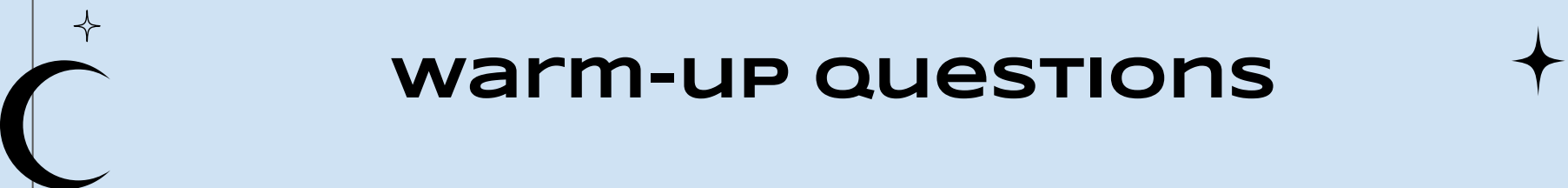


★ 02 ★

Warm-Up

Let's do a little review





warm-up questions

Please answer these on a piece of paper:

- 1) Define the following commands:
 - a) pwd,
 - b) ls,
 - c) cd,
 - d) mkdir,
 - e) nano.
 - 2) Define the following data types: string, integer, float.
 - 3) What are the two ways to run python code on the terminal?
 -
- 



warm-UP QUESTIONS Answers



- 1) `pwd` = print current working directory
`ls` = list files and folders in current working directory
`cd` = change directories
`mkdir` = make a new directory
`nano` = opens a text file editor
- 2) `string` = text/language, `integer` = whole numbers, `floats` = fractional numbers
- 3) Call: `python3 <script>.py` or call `python3` and then type python code directly on the command line

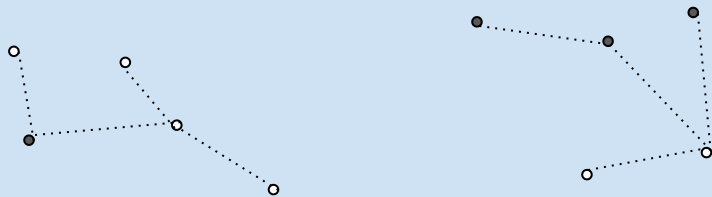
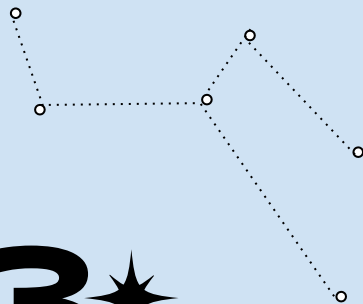




★ 03 ★

Demo

GitHub: version control





PRELUDE TO TODAY'S DEMO

Disclaimer:

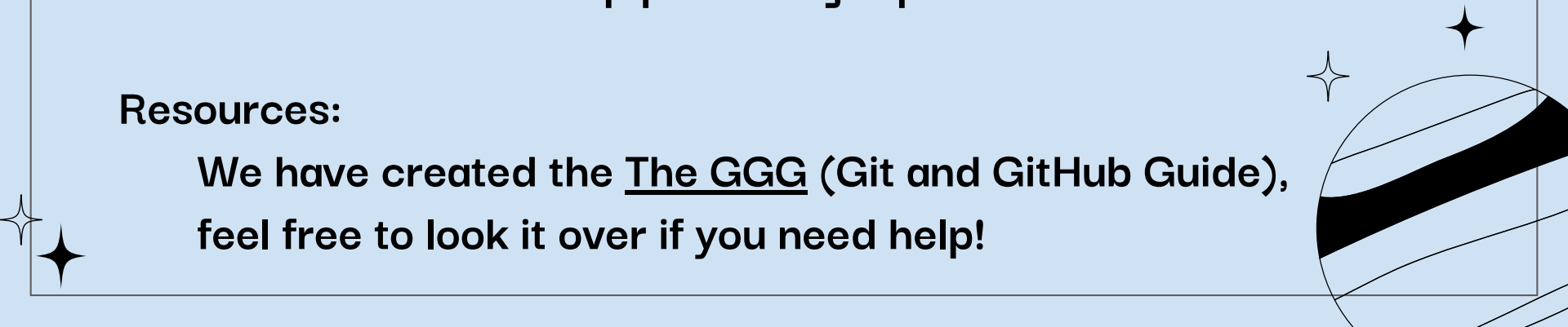
Working with Git and GitHub can be hard!

It is difficult to learn and not intuitive at the beginning.

Best advice:

Stick with it and keep practicing! I promise it will make sense

Resources:



We have created the The GGG (Git and GitHub Guide),
feel free to look it over if you need help!



BACKGROUND INFORMATION

To save our work we need to follow 3 important steps with git:

- Add our work to the staging area
- Commit (save) our work locally
- Push (save) our work remotely

Local = on your computer

Remote = on the internet





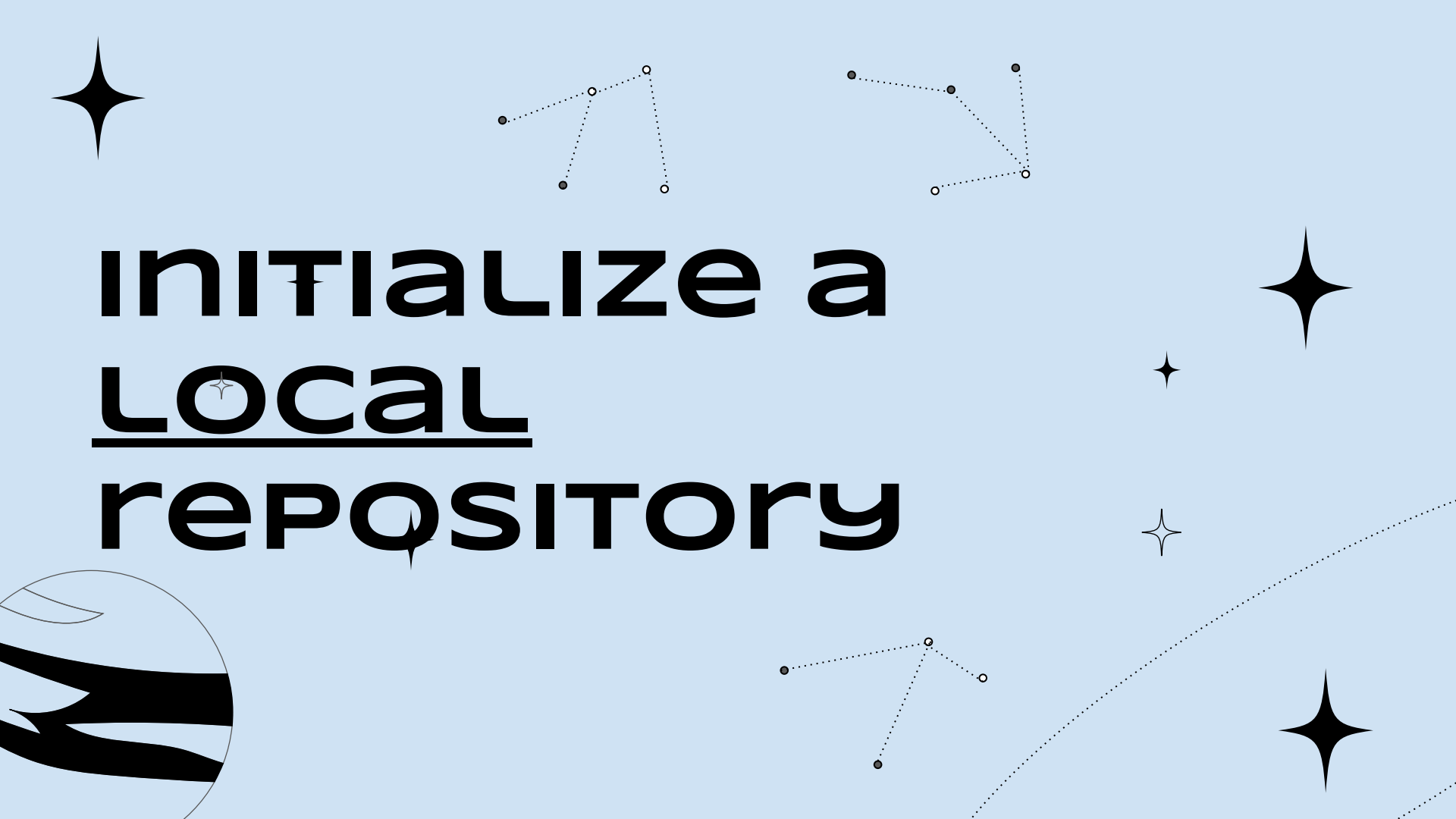
BACKGROUND INFORMATION



We will be covering A LOT of detail today, some things are super important for you to remember

- The detail is for your future reference
- Check out The GGG if you need a reference
- It's good to know the ins and outs of Git for debugging

At the end of the demo, I will tell you what the key takeaways from today should be



INITIALIZE a LOCAL repository



Background Information



The steps below are NOT ones you need to do every single time you want to save your work. These only need to happen once.

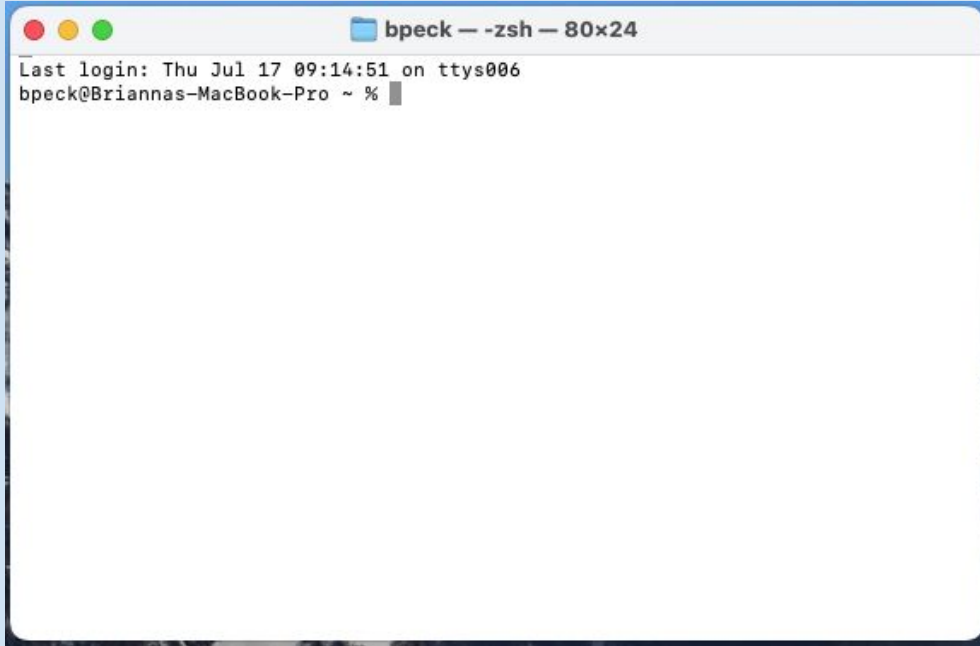
Next class, we will spend time practicing just saving our work in general



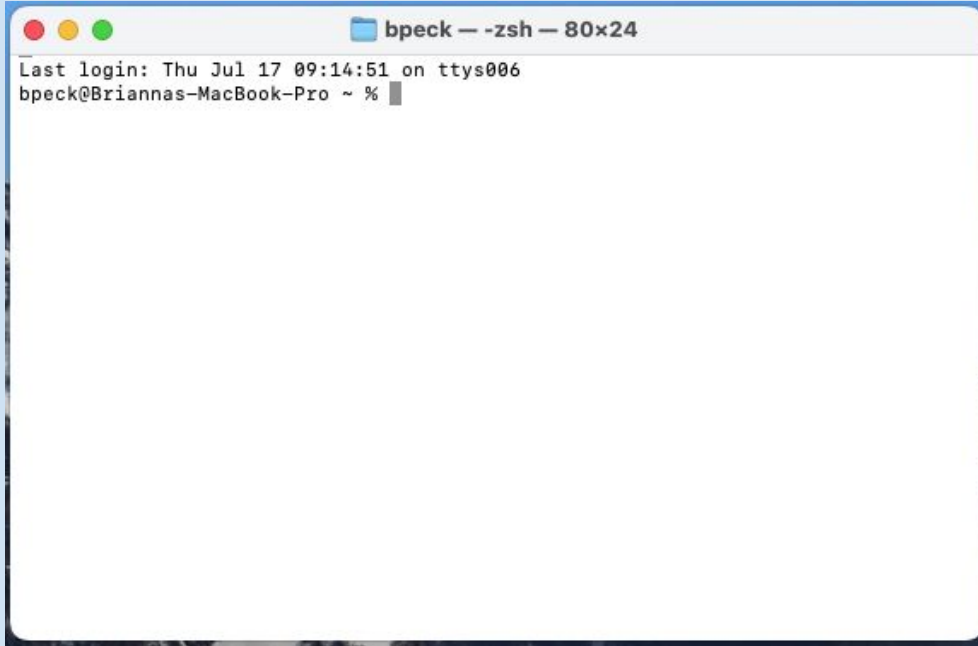
OPEN THE Terminal

Open your terminal

Windows Users: Make sure you are using GitBash as your shell and NOT PowerShell!



OPEN THE Terminal

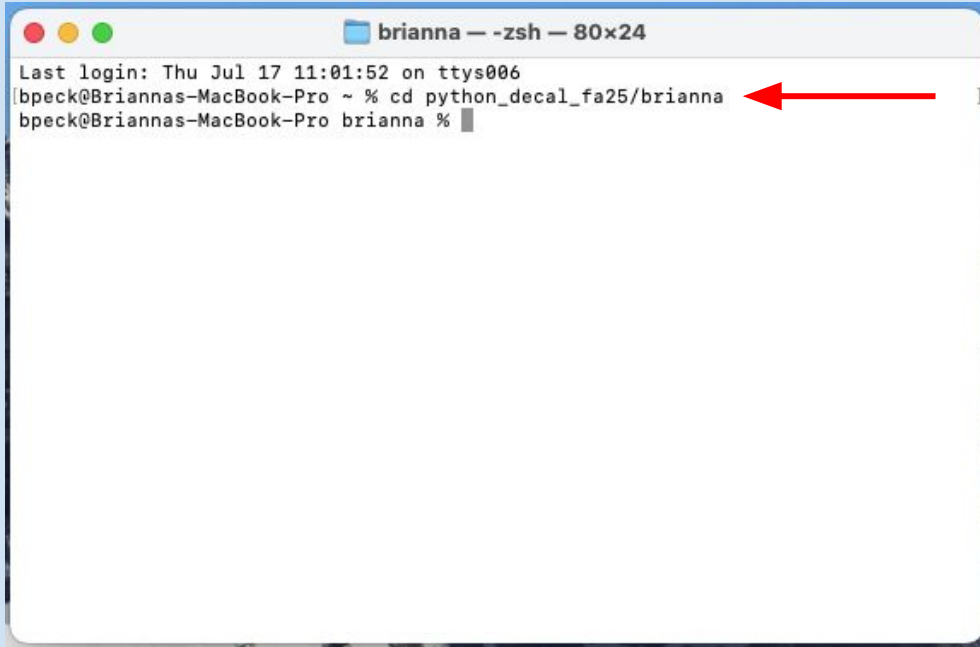
A screenshot of a macOS Terminal window. The title bar shows a blue icon, the text 'bpeck — zsh — 80x24', and three window control buttons (red, yellow, green). The terminal content shows 'Last login: Thu Jul 17 09:14:51 on ttys006' and the prompt 'bpeck@Briannas-MacBook-Pro ~ %' with a cursor. The window has a blue border and is set against a light blue background.

```
bpeck — zsh — 80x24
Last login: Thu Jul 17 09:14:51 on ttys006
bpeck@Briannas-MacBook-Pro ~ %
```

Recall: last class (and as a part of homework 1), you made a folder called yourname

Please make sure you have followed those directions before continuing

change directories

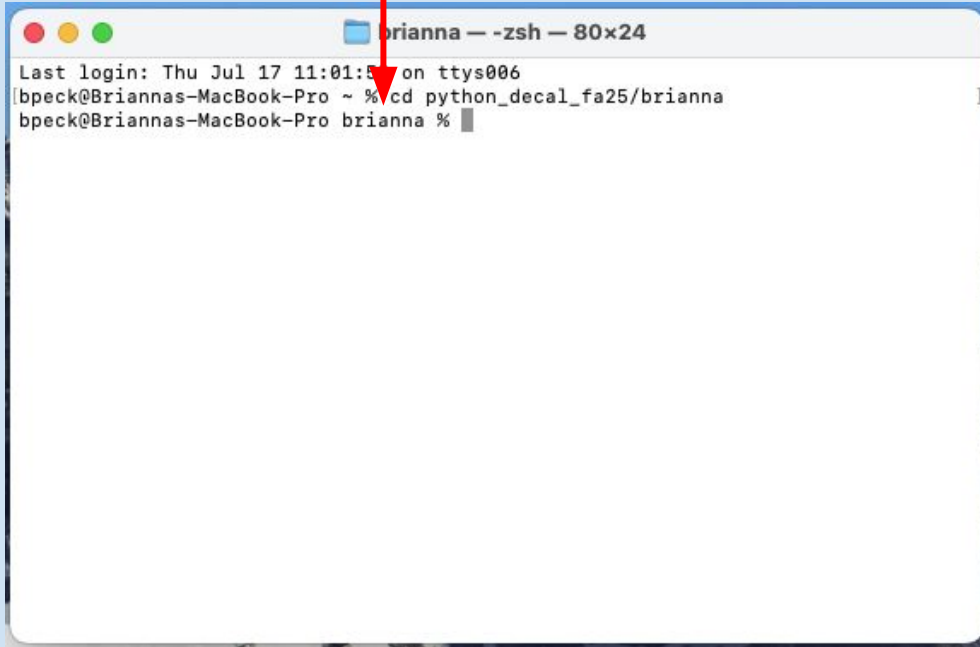
A screenshot of a macOS terminal window titled "brianna — zsh — 80x24". The window shows the command history and the current command being entered. The first line is "Last login: Thu Jul 17 11:01:52 on ttys006". The second line is "bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna", with a red arrow pointing to the path "python_decal_fa25/brianna". The third line is "bpeck@Briannas-MacBook-Pro brianna %", with a cursor at the end of the line.

```
brianna — zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna %
```

Move into your
yourname folder by
writing a file path

My name is Brianna, so
my folder is called
brianna :)

change directories

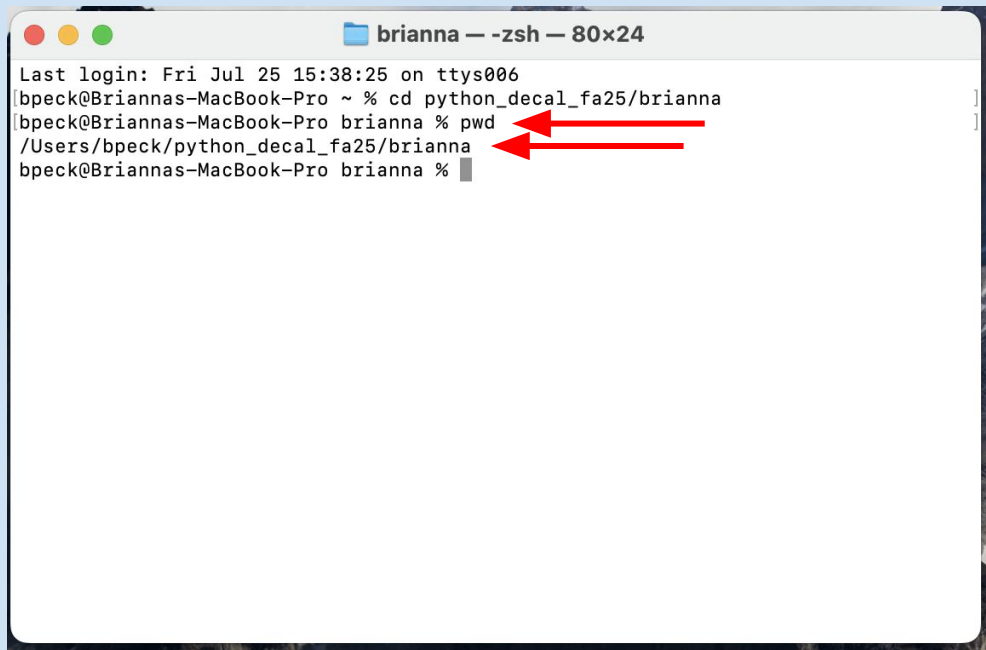
A screenshot of a macOS terminal window titled 'brianna - zsh - 80x24'. The window shows the command 'cd python_decal_fa25/brianna' being entered and executed. A red arrow points to the 'cd' command. The terminal output shows the user's location changing from '~' to 'brianna %'.

```
Last login: Thu Jul 17 11:01:5 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna %
```

Your directory may tell you
your current working
directory

If it doesn't, no worries

PRINT WORKING DIRECTORY



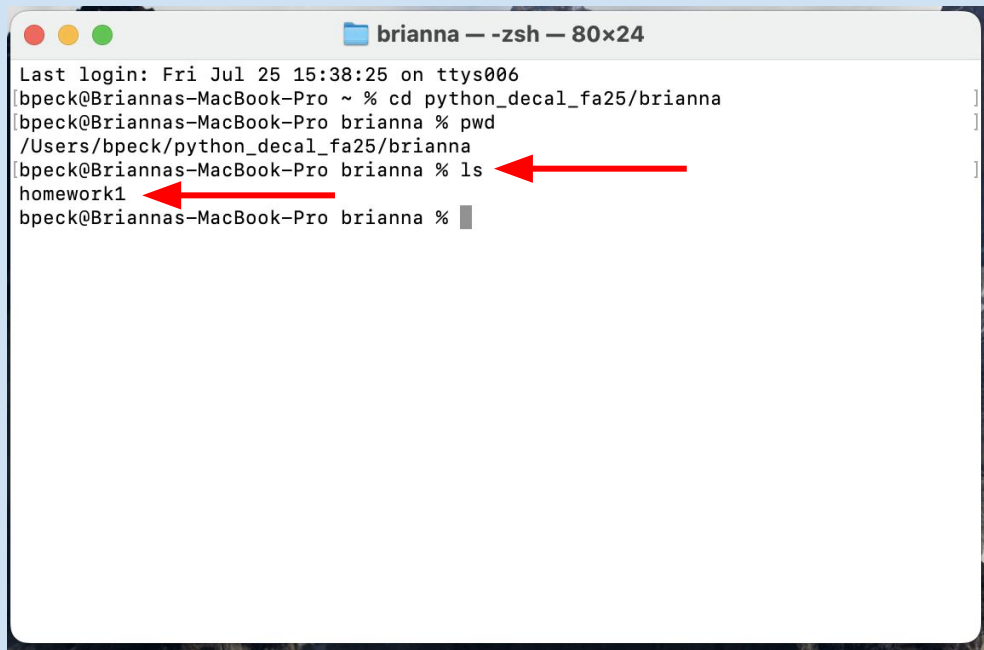
```
brianna — -zsh — 80x24
Last login: Fri Jul 25 15:38:25 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna %
```

The terminal window shows the command `pwd` being executed in the directory `python_decal_fa25/brianna`. The output is `/Users/bpeck/python_decal_fa25/brianna`. Two red arrows point from the text in the adjacent paragraph to the output of the `pwd` command.

But just to be safe, let's call pwd which will print our current working directory

We are in the folder yourname, which is in the folder python_decal_fa25

LIST THE CONTENTS

A terminal window titled 'brianna — -zsh — 80x24' showing a series of commands and their outputs. The commands are: 'cd python_decad_fa25/brianna', 'pwd', and 'ls'. The outputs are: '~ % cd python_decad_fa25/brianna', '/Users/bpeck/python_decad_fa25/brianna', and 'homework1'. Two red arrows point to the 'ls' command and its output 'homework1'.

```
brianna — -zsh — 80x24
Last login: Fri Jul 25 15:38:25 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna %
```

Next, let's list the contents of the directory

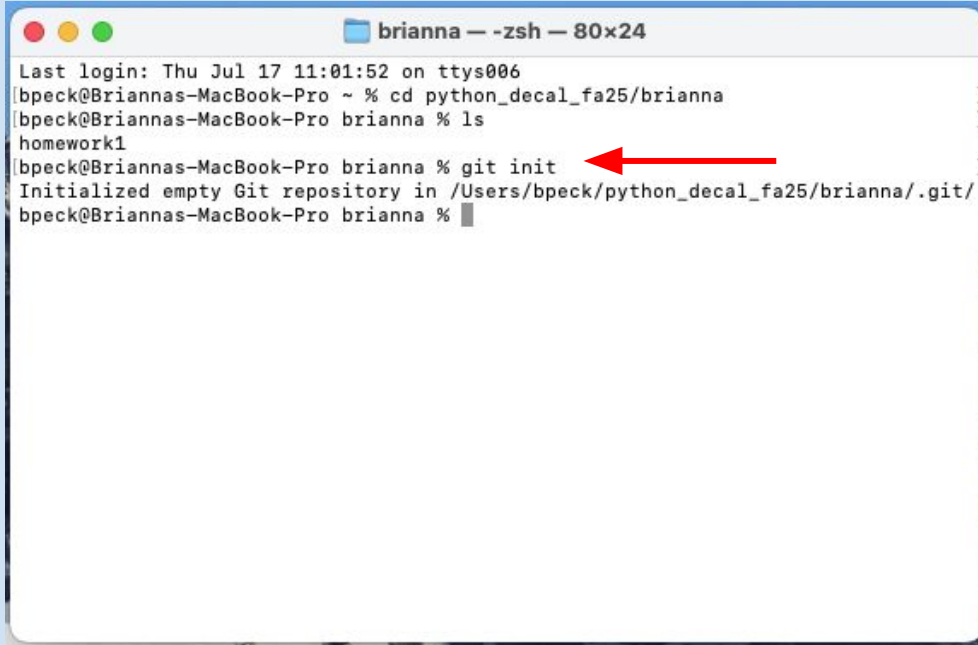
If you have been working on homework 1, then you should have a folder called homework1/

INITIALIZE a repository

To create a repository out of our directory, type:

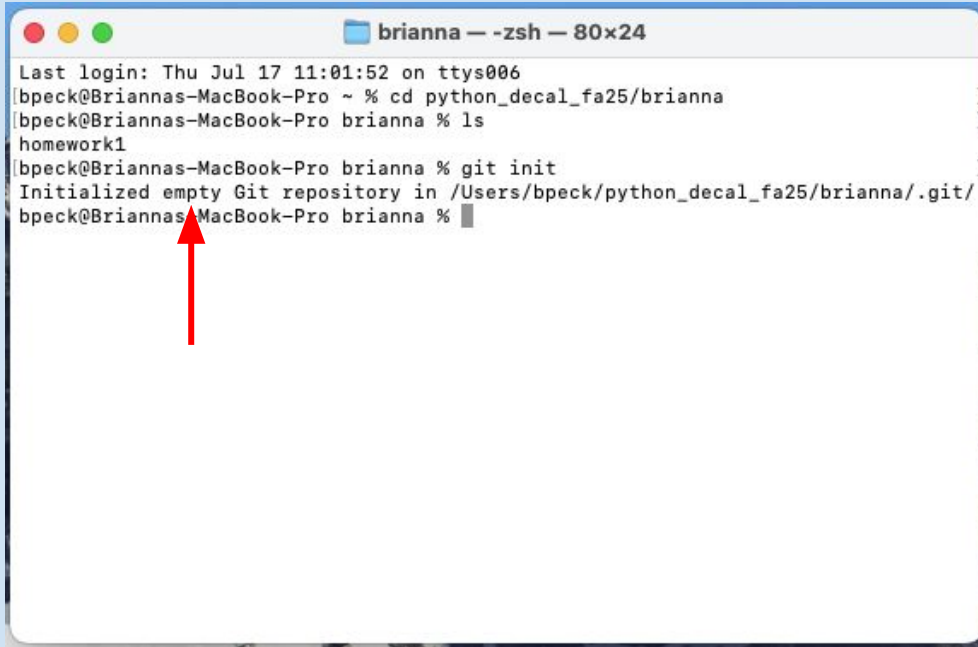
git init

Repository = a directory that has version control abilities!



```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna %
```

INITIALIZE a repository

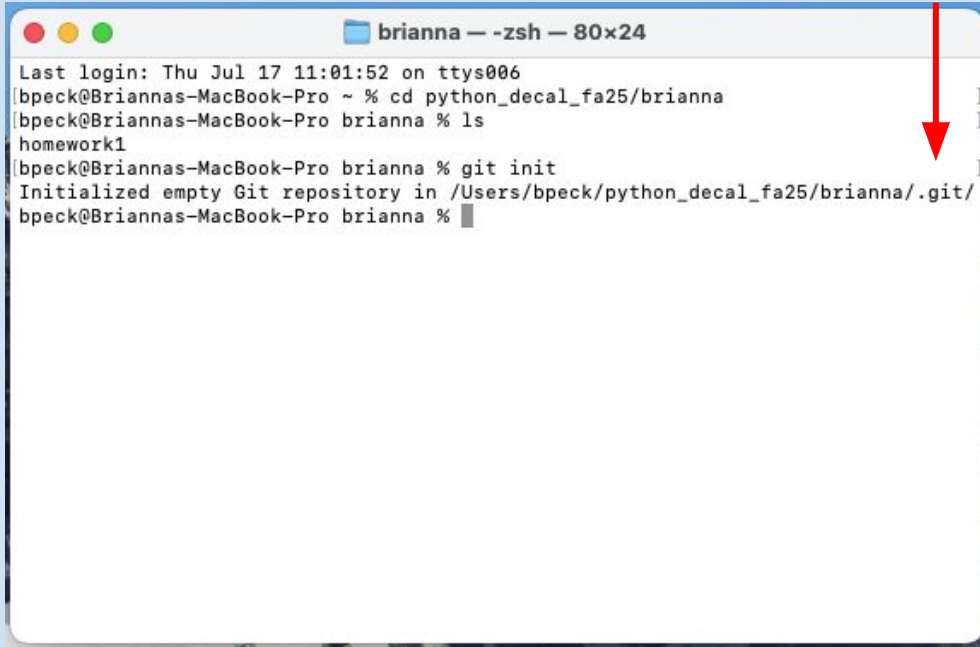
A terminal window titled 'brianna — -zsh — 80x24' showing a series of commands and their outputs. The commands are: 'cd python_decal_fa25/brianna', 'ls', and 'git init'. The outputs are: 'Last login: Thu Jul 17 11:01:52 on ttys006', 'bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna', 'bpeck@Briannas-MacBook-Pro brianna % ls', 'homework1', 'bpeck@Briannas-MacBook-Pro brianna % git init', and 'Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/'. A red arrow points to the prompt 'bpeck@Briannas-MacBook-Pro brianna %' on the last line.

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna %
```

When we look at the message that popped up, we see that our repository is “empty”

That means we don’t have any saved changes yet, we need to manually save our changes every single time

HIDDEN GIT FOLDER



```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna %
```

When we look at the filepath, we see this new folder: .git/

This is actually a hidden folder and won't appear if we just call ls

It holds the information for our version control

LIST NON-HIDDEN CONTENTS

So when we call ls we don't see that .git/ folder, we only see the original homework1/

But if we call ls -a, which shows hidden files, we will see it

```
brianna — -zsh — 80x24
Last login: Fri Jul 25 15:38:25 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna %
```

LIST HIDDEN CONTENTS

```
brianna — -zsh — 80x24
Last login: Fri Jul 25 15:38:25 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .DS_Store  .git      homework1
bpeck@Briannas-MacBook-Pro brianna %
```

The hidden .git/ folder appeared, it will store information in the background

The rest mean:

. = current directory

.. = parent directory

.DS_Store = created by the mac Finder, irrelevant

CALL GIT STATUS

Let's call a command that is really helpful for debugging

git status

It is a command that tells you the “status” of your repository

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

main is our Branch

Let's work through
line-by-line what it has told
us

On branch main

We haven't covered
branches, but this tells us
the name of the branch we
are on, so main

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/
nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

AN EMPTY repository

No commits yet

This means we have saved no changes yet, we are NOT performing version control yet

So our repository is still acting like a directory

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet ←

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

GIT sees CHANGES

Untracked files:

This means git sees that there are files in the directory that we haven't backed up yet

Secret lecture code: save

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

GIT TELLS US WHAT IT SEES

(use “git add <file>...” ...)

This line, is git telling us
what to do to save our work

We will call git add shortly

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```


GIT TELLS US HOW TO save work ✨

homework1/

This is git telling us that we have a folder called homework1, that has NOT been saved

git does NOT save work automatically

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```


**saving your
code**



BACKGROUND INFORMATION



As we go through these steps, your terminal might give you slightly different responses

- Especially if you have NOT started homework 1
- This lecture assumes that you have started creating your *homework1.py* Python file

GIT TELLS US HOW TO save work

nothing added to commit...

This message is once again
telling us we need to call:
git add, to save our work

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

GIT TELLS US HOW TO save work ✨

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna %
```

There are two main ways to call git add

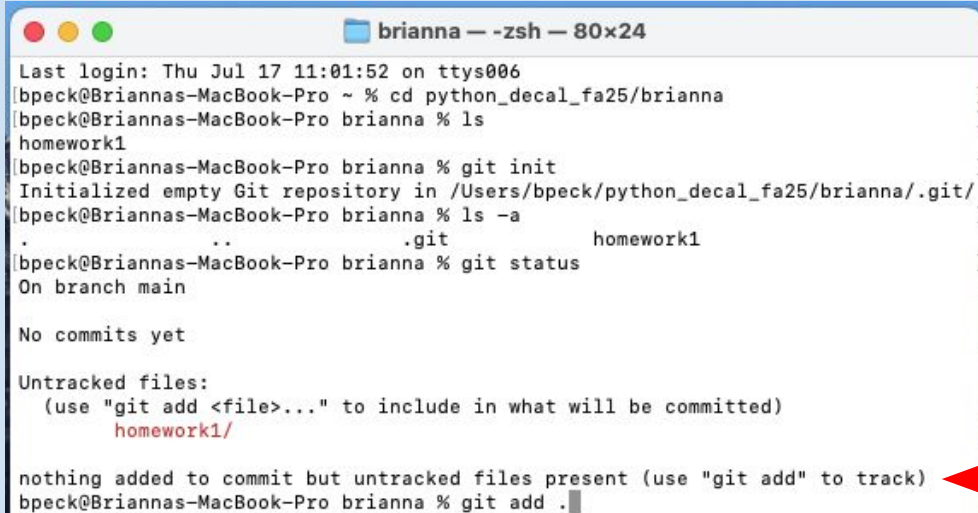
1) git add .

This saves ALL your files

2) git add <file>

This saves only the files you specify

ADD TO STAGING area



```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
```

For now, we want to save everything in this repository so let's call git add .

Now git will search through all of yourname/ and looks for files to save

But we are not done yet, there are more steps

ADDED TO STAGING area

Once you have called:
git add .

Nothing will appear! Git is now “tracking” these files and they are in a “staging” area

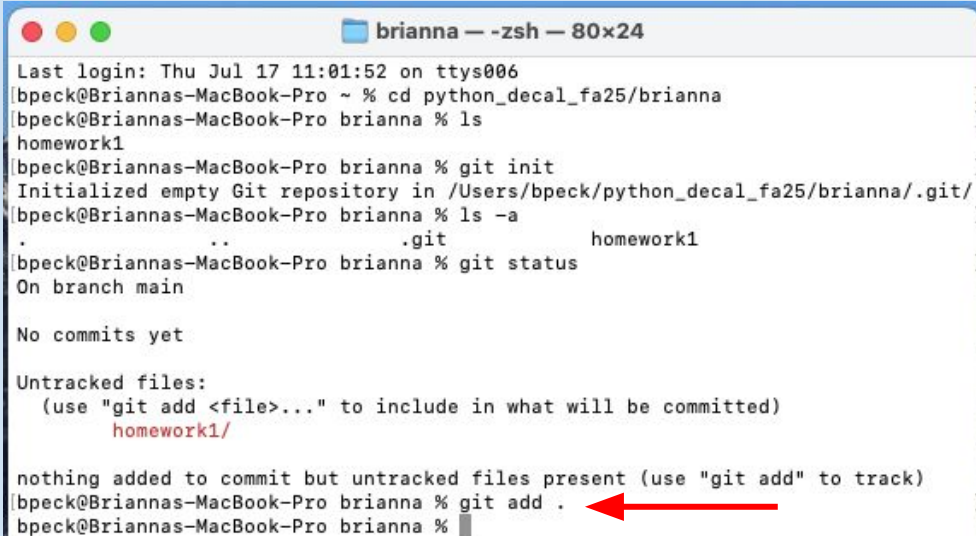
```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna %
```

ADDED TO STAGING area



```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:01:52 on ttys006
bpeck@Briannas-MacBook-Pro ~ % cd python_decal_fa25/brianna
bpeck@Briannas-MacBook-Pro brianna % ls
homework1
bpeck@Briannas-MacBook-Pro brianna % git init
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna %
```

Tracking = Git knows about the existence of a file and is keep a history of its changes

Staging Area (rough draft zone) = a place to make last-minute changes before committing files to your saved history

GIT STATUS GIVES US AN UPDATE

```
brianna — -zsh — 80x24
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.
..
.git
homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   homework1/homework1.py
bpeck@Briannas-MacBook-Pro brianna %
```

Let's call:

git status again

We see these lines again:

- On branch main
- No commits yet

Meaning, we haven't
changed branches or
completely saved our files

STUFF IN STAGING area

Changes to be committed:

This means we have added files to our staging area (rough draft zone)

If we want to unadd files, this would be the time to do it

```
brianna — -zsh — 80x24
Initialized empty Git repository in /Users/bpeck/python_decad_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.
..
.git
homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   homework1/homework1.py
bpeck@Briannas-MacBook-Pro brianna %
```

HOW TO DELETE FROM AREA

(use “git rm --cached <file...”

For example, if we added a multiple versions of a file, and only want to keep one of them

We could call git rm to remove them from the staging area

```
brianna — -zsh — 80x24
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -la
.
..
.git
homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   homework1/homework1.py
```

HOW TO DELETE FROM AREA

```
brianna — -zsh — 80x24
Initialized empty Git repository in /Users/bpeck/python_decal_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
      new file:   homework1/homework1.py
bpeck@Briannas-MacBook-Pro brianna %
```

But we want to just save the changes on ALL of our files right now

So we won't be calling git rm

WHAT FILE IS IN THE Area

new file: homework1/hom...

This line tells us that a new file has been added to the staging area

Git is awaiting us to fully save its progress

```
brianna — -zsh — 80x24
Initialized empty Git repository in /Users/bpeck/python_decad_fa25/brianna/.git/
bpeck@Briannas-MacBook-Pro brianna % ls -a
.      ..      .git      homework1
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

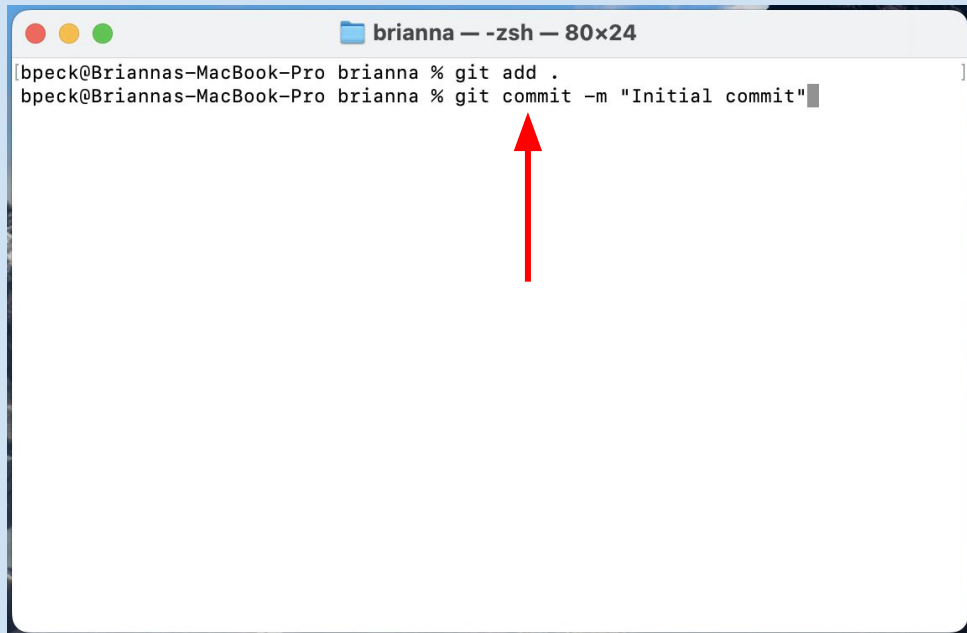
Untracked files:
  (use "git add <file>..." to include in what will be committed)
      homework1/

nothing added to commit but untracked files present (use "git add" to track)
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
      new file:   homework1/homework1.py
```

saving work TO LOCAL COMP.

A terminal window titled 'brianna — -zsh — 80x24' showing two commands: 'git add .' and 'git commit -m "Initial commit"'. A red arrow points to the 'commit' command.

```
brianna % git add .  
brianna % git commit -m "Initial commit"
```

The next step in the process of saving our work is to call git commit

The full command is:
git commit -m "message"

For the message you can write something like:
Saving homework1/ folder

saving work TO LOCAL COMP.

Git will give you a really long message in return, let's work through a bit of the message line-by-line again

```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

THE commit Message

[main (root-commit) b03c2ac]

main = the branch you are on
(root-commit) = the command
you just called created the
first commit of the repository
b03c2ac = the commit number
(i.e., ID #) of the version you
just saved

```
brianna -- -zsh -- 80x24
[bpeck@Briannas-MacBook-Pro brianna % git add .
[bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
6 files changed, 262 insertions(+)
create mode 100644 .DS_Store
create mode 100644 homework1/homework1.py
create mode 100644 homework1/hw1_commandline.png
create mode 100644 homework1/hw1_folder.png
create mode 100644 homework1/hw1_pwd_and_ls.png
create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```


THE COMMIT MESSAGE

Initial Commit

This is the message
associated with the version
you just saved to your
computer

```
brianna — -zsh — 80x24
[bpeck@Briannas-MacBook-Pro brianna % git add .
[bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit ←
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

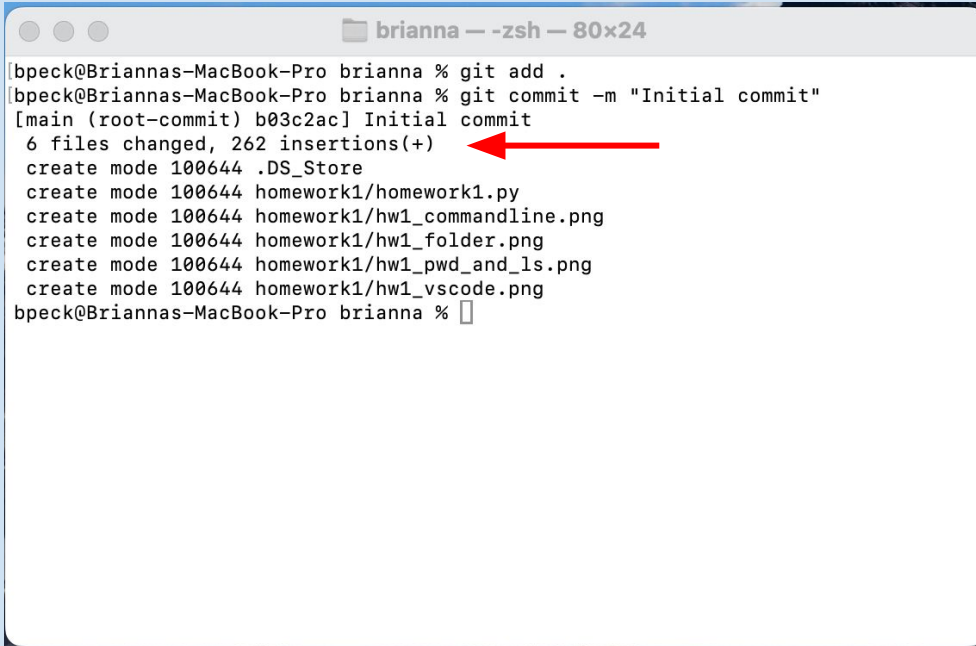
THE COMMIT MESSAGE

```
brianna -- -zsh -- 80x24
[bpeck@Briannas-MacBook-Pro brianna % git add .
[bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
6 files changed, 262 insertions(+)
create mode 100644 .DS_Store
create mode 100644 homework1/homework1.py
create mode 100644 homework1/hw1_commandline.png
create mode 100644 homework1/hw1_folder.png
create mode 100644 homework1/hw1_pwd_and_ls.png
create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

This is git telling us that we have changed six files with 262 line insertions

- So, six files are being saved for the first time
- In all of those files, 262 lines of code have been saved

THE COMMIT MESSAGE



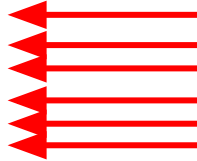
```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
6 files changed, 262 insertions(+)
create mode 100644 .DS_Store
create mode 100644 homework1/homework1.py
create mode 100644 homework1/hw1_commandline.png
create mode 100644 homework1/hw1_folder.png
create mode 100644 homework1/hw1_pwd_and_ls.png
create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```

Since this is the first time we are saving changes, git has nothing to compare the changes against

So this line just tells us what is currently in our repository prior to saving our work

THE COMMIT MESSAGE

```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
6 files changed, 262 insertions(+)
create mode 100644 .DS_Store
create mode 100644 homework1/homework1.py
create mode 100644 homework1/hw1_commandline.png
create mode 100644 homework1/hw1_folder.png
create mode 100644 homework1/hw1_pwd_and_ls.png
create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna %
```



All of these lines means different files that are being saved for the first time in our version history

If you have completed homework 1 this is what your repository would look like

CALLING GIT STATUS FOR UPDATE

Let's call git status again

We see the original:

- On branch main

But now we see:

- Nothing to commit,
working tree clean
- So Git has properly saved
all of our work!

```
brianna — -zsh — 80x24
[bpeck@Briannas-MacBook-Pro brianna % git add .]
[bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"]
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
[bpeck@Briannas-MacBook-Pro brianna % git status]
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna %
```

CALL GIT LOG FOR HISTORY

The command git log is also pretty helpful, it tells us the actual history of our folder

```
brianna — -zsh — 80x24
[bpeck@Briannas-MacBook-Pro brianna % git add .]
[bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"]
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
[bpeck@Briannas-MacBook-Pro brianna % git status]
On branch main
nothing to commit, working tree clean
[bpeck@Briannas-MacBook-Pro brianna % git log]
```



CALL GIT LOG FOR HISTORY

```
brianna — -zsh — 80x24
[bpeck@Briannas-MacBook-Pro brianna % git add .]
[bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"]
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
[bpeck@Briannas-MacBook-Pro brianna % git status]
On branch main
nothing to commit, working tree clean
[bpeck@Briannas-MacBook-Pro brianna % git log]
commit b03c2ac3285a925df67386af3c7896ed6b74fd53 (HEAD -> main)
Author: Brianna Peck <bpeck114@berkeley.edu>
Date: Sat Jul 26 08:17:24 2025 -1000

    Initial commit
bpeck@Briannas-MacBook-Pro brianna %
```

There is our commit number again, but this time the full commit number

(HEAD -> main):

HEAD = the current commit number you are using

-> main = you are using the latest commit on your main branch

CALL GIT LOG FOR HISTORY

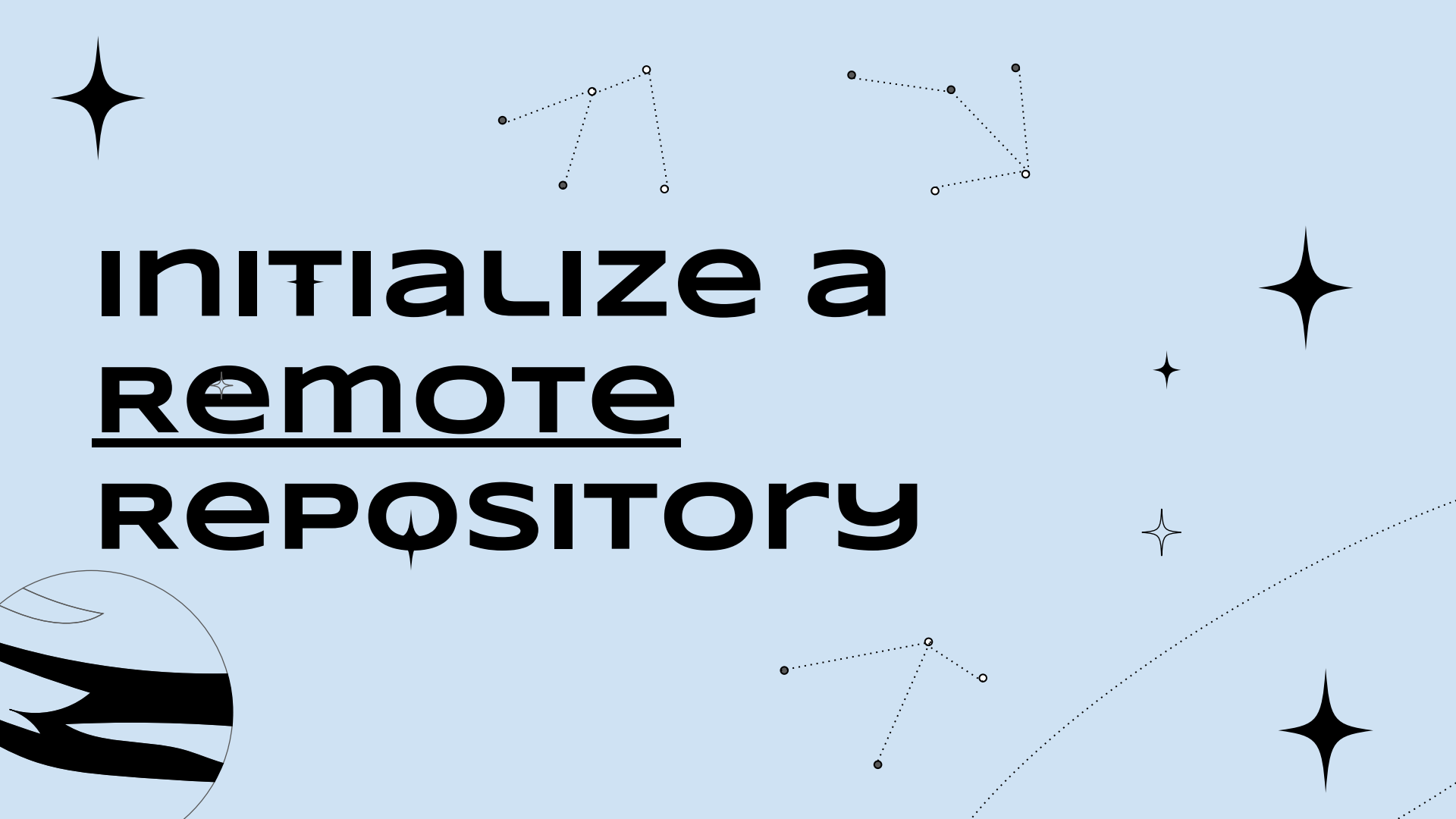
The person who made the commit and their associated email address

The date and time the commit was saved

The message of the commit

```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Initial commit"
[main (root-commit) b03c2ac] Initial commit
 6 files changed, 262 insertions(+)
 create mode 100644 .DS_Store
 create mode 100644 homework1/homework1.py
 create mode 100644 homework1/hw1_commandline.png
 create mode 100644 homework1/hw1_folder.png
 create mode 100644 homework1/hw1_pwd_and_ls.png
 create mode 100644 homework1/hw1_vscode.png
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git log
commit b03c2ac3285a925df67386af3c7896ed6b74fd53 (HEAD -> main)
Author: Brianna Peck <bpeck114@berkeley.edu>
Date: Sat Jul 26 08:17:24 2025 -1000

    Initial commit
bpeck@Briannas-MacBook-Pro brianna %
```

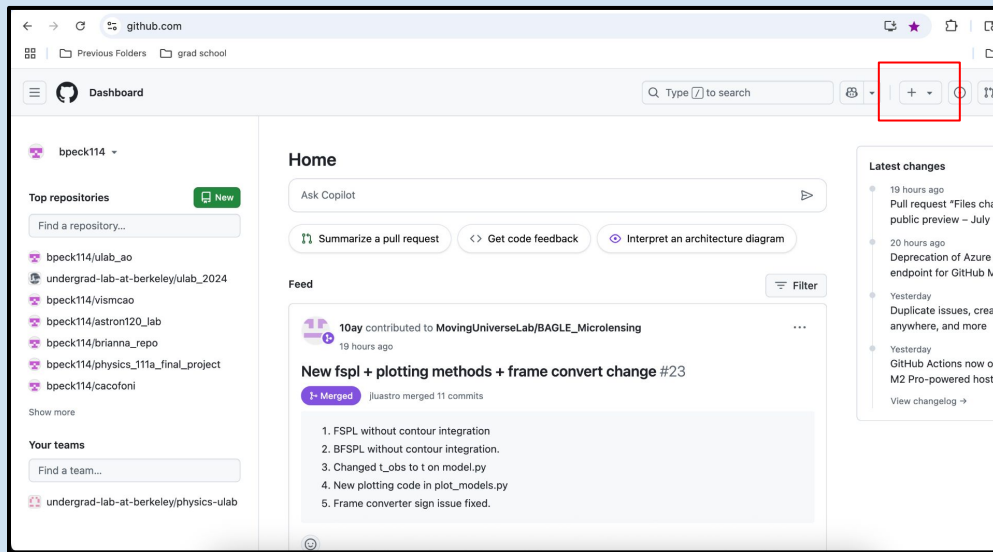



INITIALIZE a REMOTE REPOSITORY

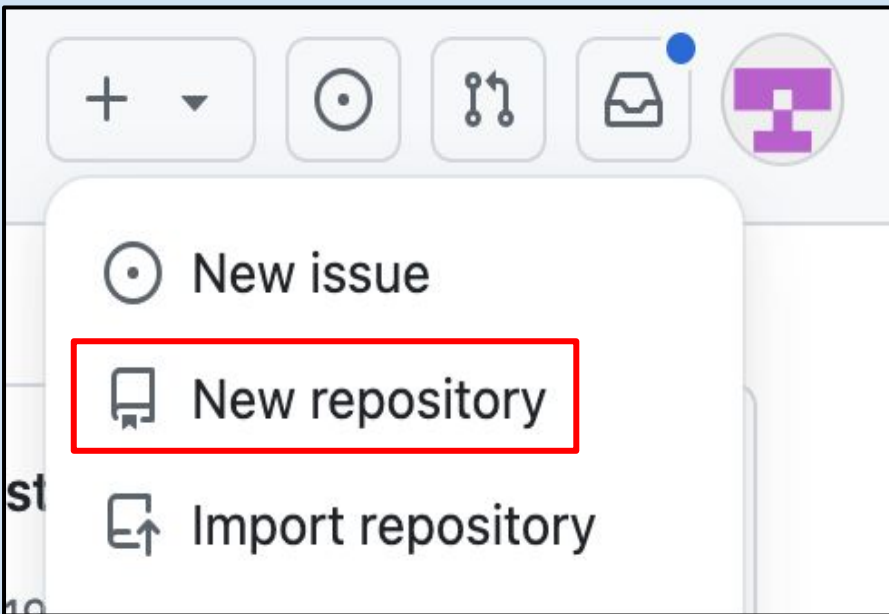
OPEN GITHUB

On a web browser, open up GitHub. Sign in with your @berkeley.edu email address.

On the top right-hand corner, you will see a box with a plus sign. Click on that plus sign.



make a new repository



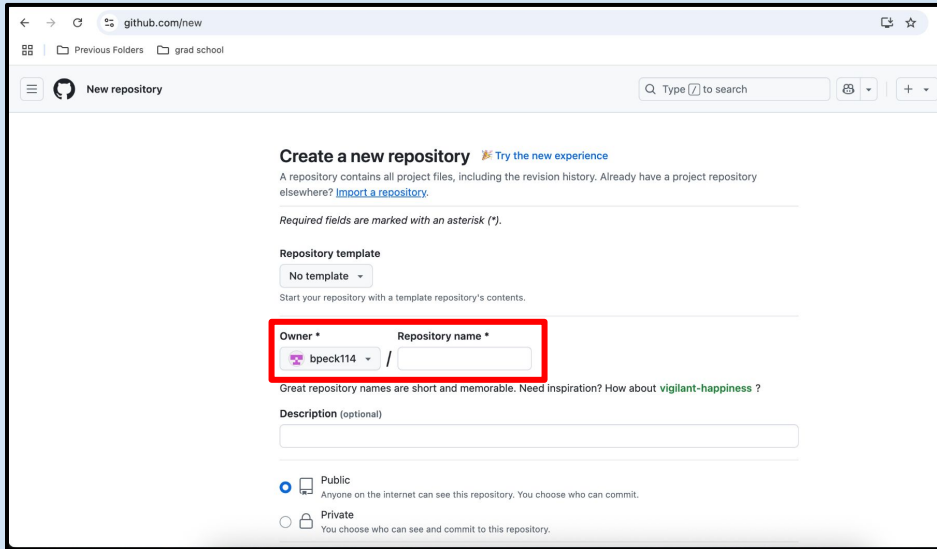
A drop-down menu will appear.

Click on the tab that says:
New repository

create a new repository

Another new page will open
titled: Create a new repository

Go to the section called:
Repository name



github.com/new

Previous Folders grad school

New repository

Q Type [Z] to search

Create a new repository [Try the new experience](#)

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

Repository template

No template

Start your repository with a template repository's contents.

Owner * Repository name *

bpeck114 /

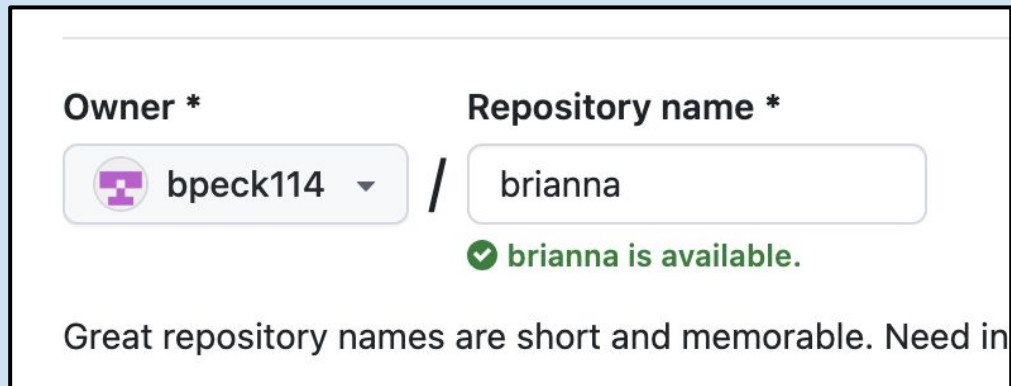
Great repository names are short and memorable. Need inspiration? How about [vigilant-happiness](#) ?

Description (optional)

☒ Public
Anyone on the internet can see this repository. You choose who can commit.

☐ Private
You choose who can see and commit to this repository.

same Name as LOCAL REPO



A screenshot of the GitHub repository creation interface. It features two input fields: 'Owner *' with a dropdown menu showing 'bpeck114' and a 'Repository name *' text box containing 'brianna'. Below the repository name field, a green checkmark icon is followed by the text 'brianna is available.'. At the bottom of the form, a note reads: 'Great repository names are short and memorable. Need in'.

Owner * Repository name *

bpeck114 / brianna

✓ brianna is available.

Great repository names are short and memorable. Need in

Give it the same title as what you called the repository on your computer!!

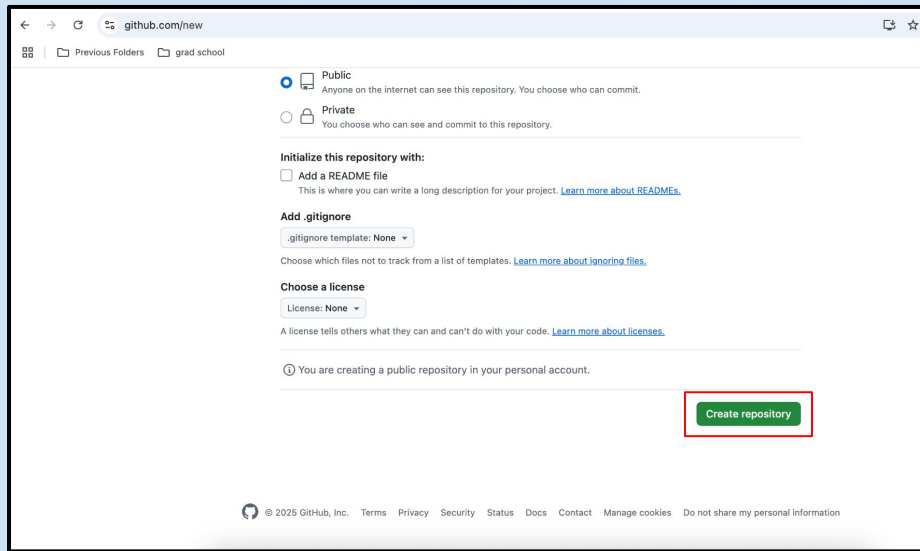
So: yourname

In my case, my name is Brianna. I called my repository brianna

create THE new repository

Leave all other settings on default

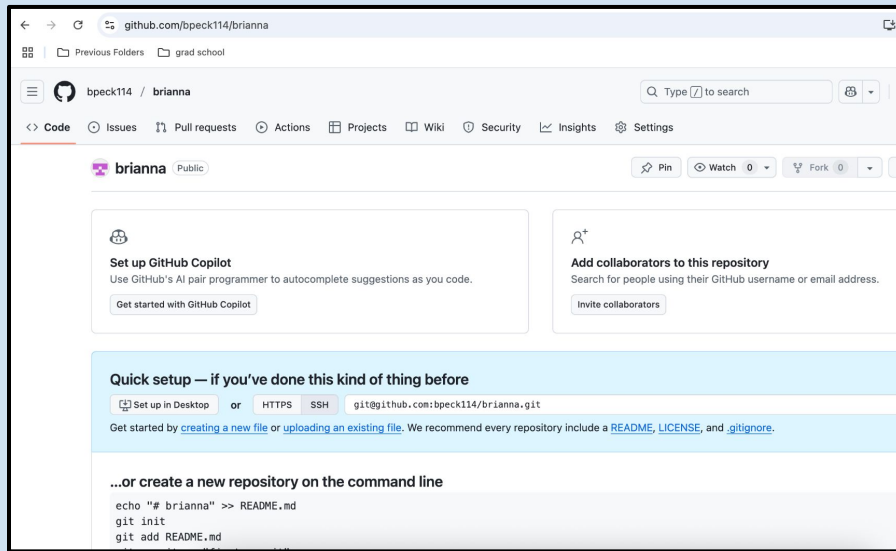
Scroll to the bottom of the page. Click on :
Create repository

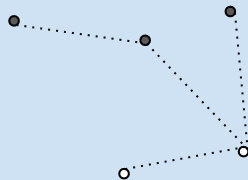
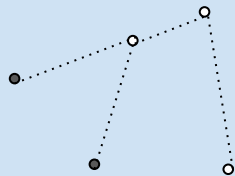


The screenshot shows the GitHub 'Create new repository' page. The browser address bar shows 'github.com/new'. The page has a light yellow background. At the top, there are navigation links: 'Previous Folders' and 'grad school'. The main content area has a white background. It starts with two radio buttons for 'Public' (selected) and 'Private'. Below this is a section 'Initialize this repository with:' with a checkbox for 'Add a README file'. Then there's a section 'Add .gitignore' with a dropdown menu set to '.gitignore template: None'. Below that is a section 'Choose a license' with a dropdown menu set to 'License: None'. At the bottom, there is a green button labeled 'Create repository' which is highlighted with a red rectangle. The footer contains copyright information and links for Terms, Privacy, Security, Status, Docs, Contact, and Manage cookies.

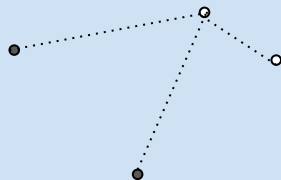
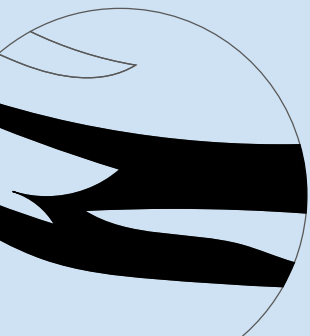
OPENS ANOTHER new WINDOW ✨

Another new window will open
after creating your remote
repository

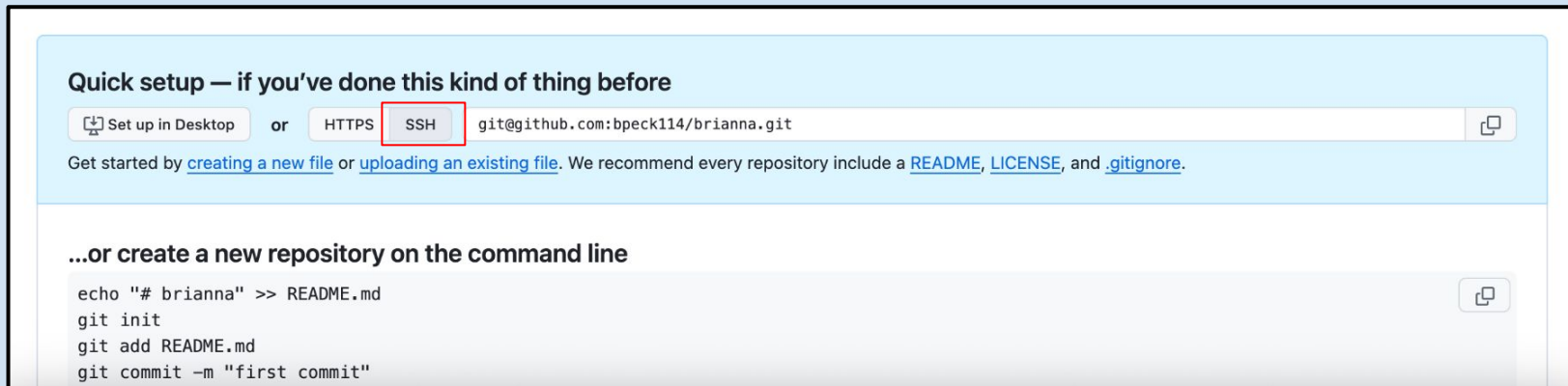




connect Local + Remote



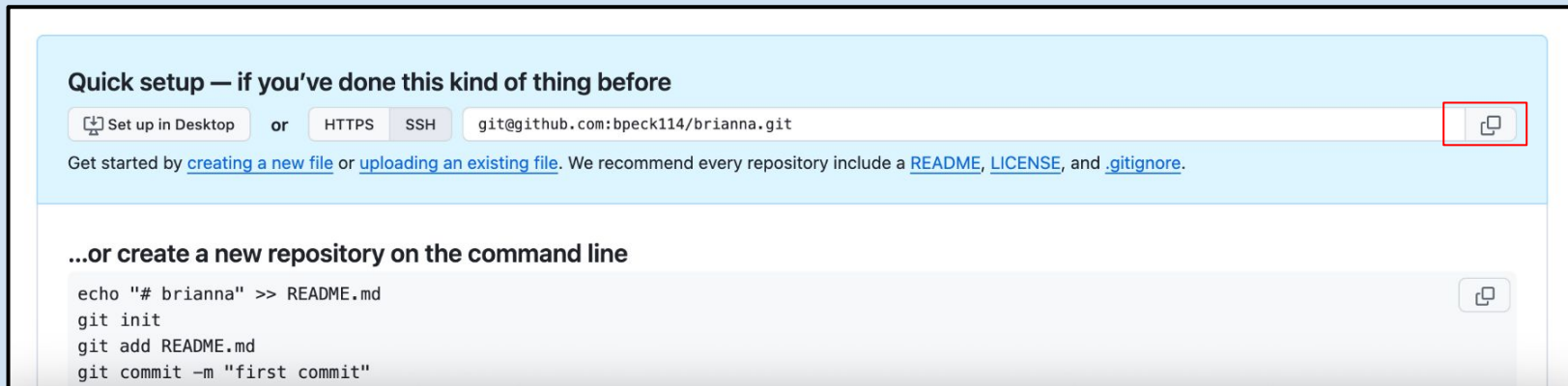
COPY THE SSH LINK, NOT HTTPS



Go to the Quick Setup section on new window

IMPORTANT!!! Make sure you have the SSH tab highlighted!

COPY THE SSH LINK, NOT HTTPS ✨



Quick setup — if you've done this kind of thing before

 Set up in Desktop or **HTTPS** **SSH** `git@github.com:bpeck114/brianna.git` 

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

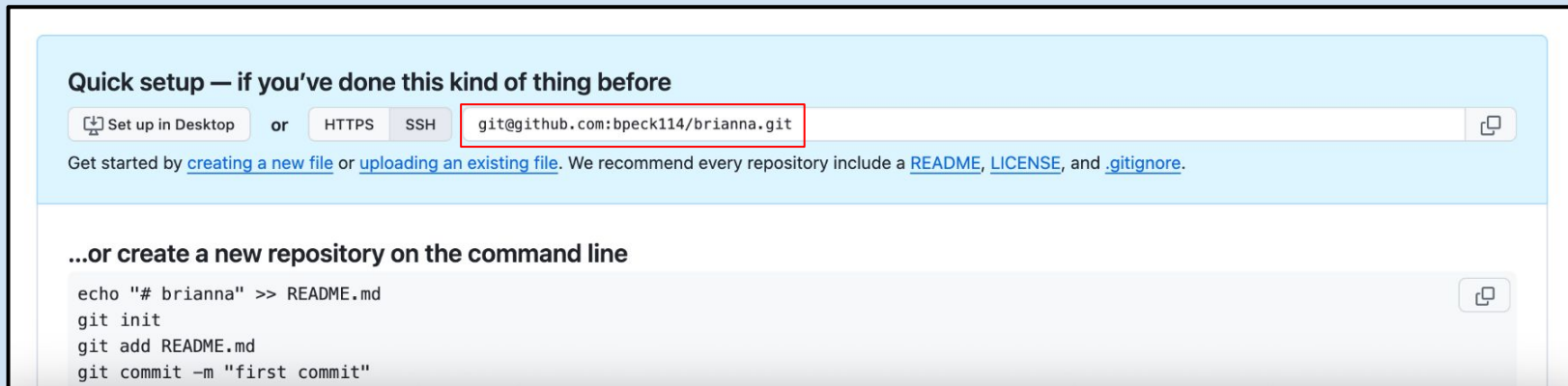
...or create a new repository on the command line

```
echo "# brianna" >> README.md
git init
git add README.md
git commit -m "first commit"
```



Click on the clipboard icon on the right-hand side of the screen to copy the link

COPY THE SSH LINK, NOT HTTPS



Quick setup — if you've done this kind of thing before

 Set up in Desktop or **HTTPS** **SSH** `git@github.com:bpeck114/brianna.git` 

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

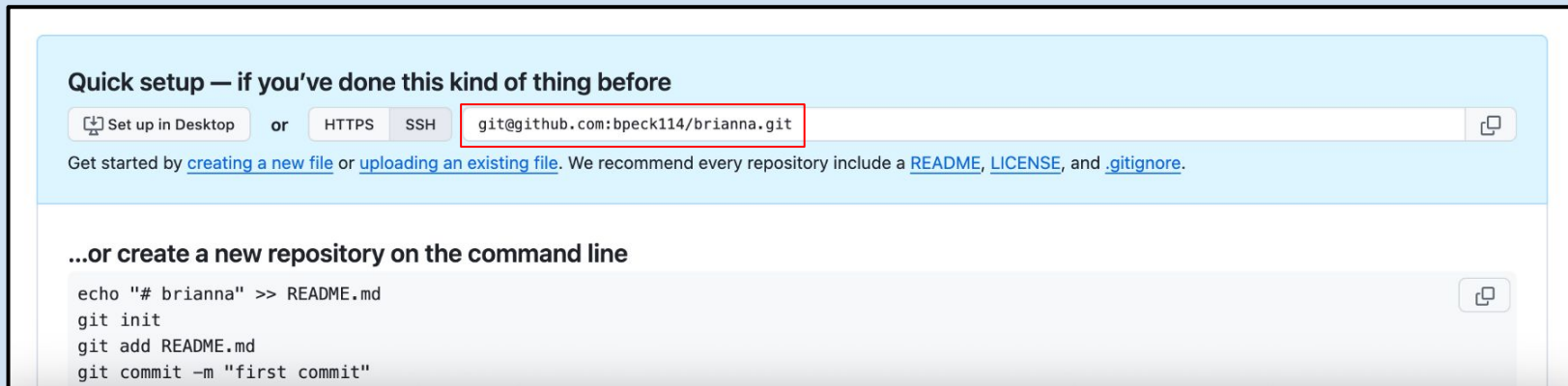
...or create a new repository on the command line

```
echo "# brianna" >> README.md
git init
git add README.md
git commit -m "first commit"
```



It happens every year, but there is always someone who copies the wrong link, make sure to copy the SSH link!

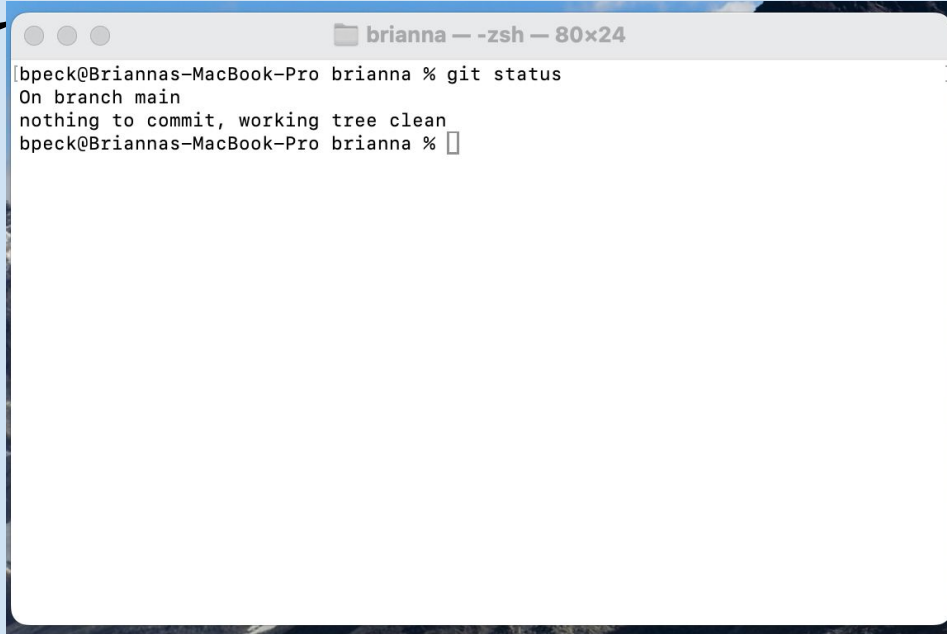
COPY THE SSH LINK, NOT HTTPS



The SSH link ends with .git,

- If the link you are copying DOES NOT end with .git then you are copying the WRONG one.

OPEN THE Terminal

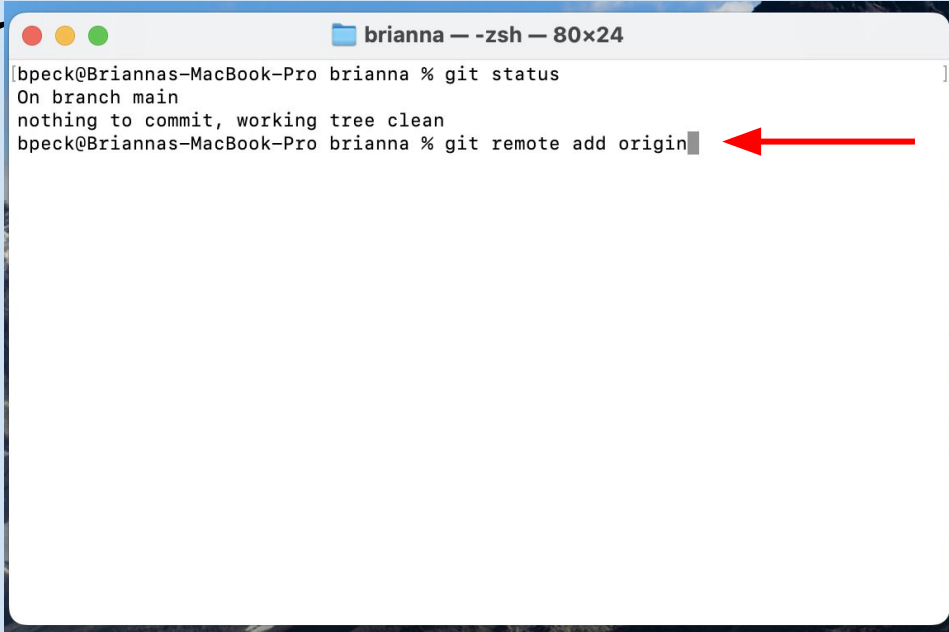
A screenshot of a terminal window titled 'brianna — zsh — 80x24'. The window shows the output of the 'git status' command. The text inside the terminal is: 'bpeck@Briannas-MacBook-Pro brianna % git status', 'On branch main', 'nothing to commit, working tree clean', and 'bpeck@Briannas-MacBook-Pro brianna %'.

```
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna %
```

Open up the terminal again

Make sure you are in the yourname repository, otherwise this next step won't work!

ADD THE REMOTE ORIGIN



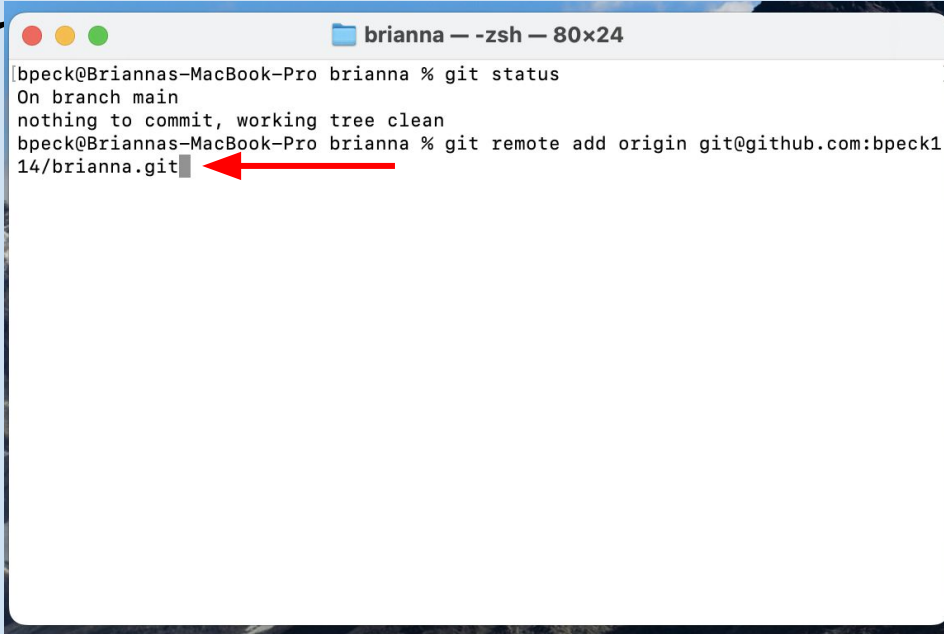
```
brianna — zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin
```

A terminal window titled "brianna — zsh — 80x24" showing the output of the `git status` command and the command `git remote add origin`. A red arrow points to the end of the command line.

DON'T Hit Return/Enter yet,
we will work through this
command in steps

Type:
git remote add origin

PASTE THE LINK YOU COPIED



```
brianna — zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck14/brianna.git
```

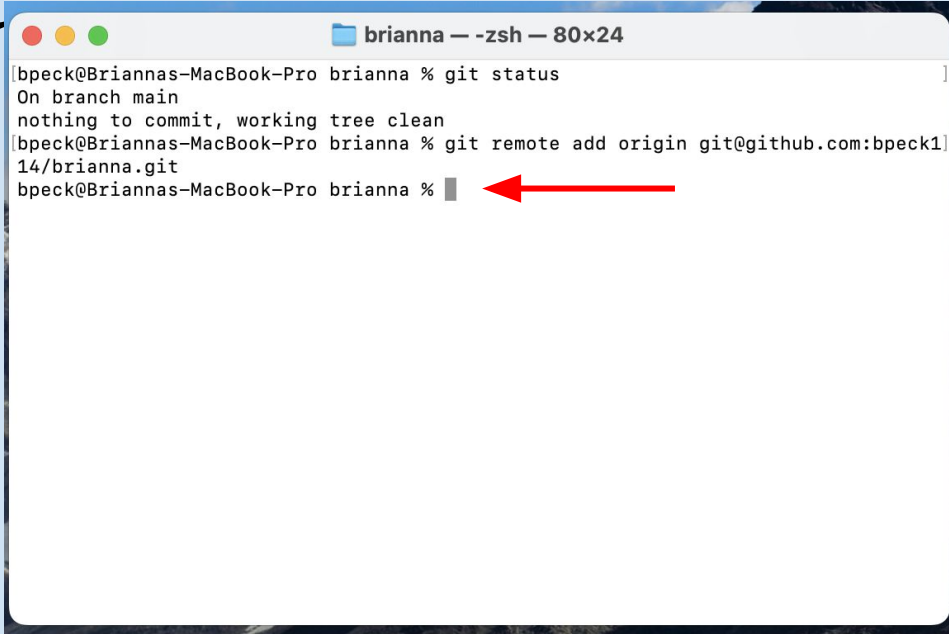
A terminal window titled "brianna — zsh — 80x24" showing the output of "git status" and the command "git remote add origin git@github.com:bpeck14/brianna.git". A red arrow points to the end of the command line.

After those lines, paste the url
you copied from GitHub

Make sure it is the SSH url!

Please, please, please double
check or you will get an error

PASTE THE LINK YOU COPIED



```
brianna — zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck14/brianna.git
bpeck@Briannas-MacBook-Pro brianna %
```

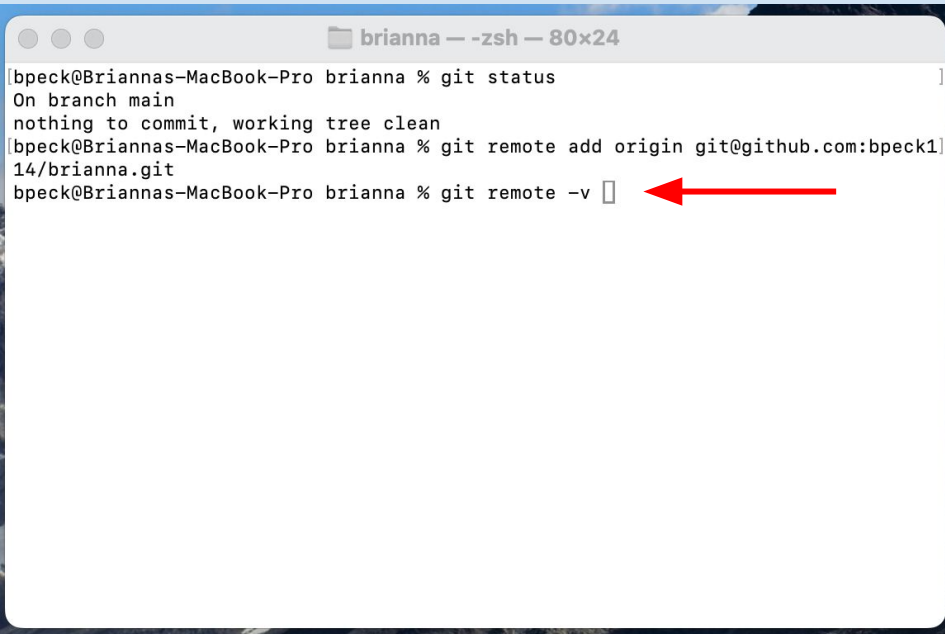
A red arrow points to the prompt character at the end of the last line.

Nothing will show up afterwards, but let's check to make sure the connection is established

CHECK THE CONNECTION

Type: git remote -v

This command will get check
the connection between our
local and remote repositories

A terminal window titled 'brianna — zsh — 80x24' is shown. It contains the following text:

```
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck1
14/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
```

A red arrow points to the prompt of the last command line.

CHECK THE CONNECTION

```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck114/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna %
```

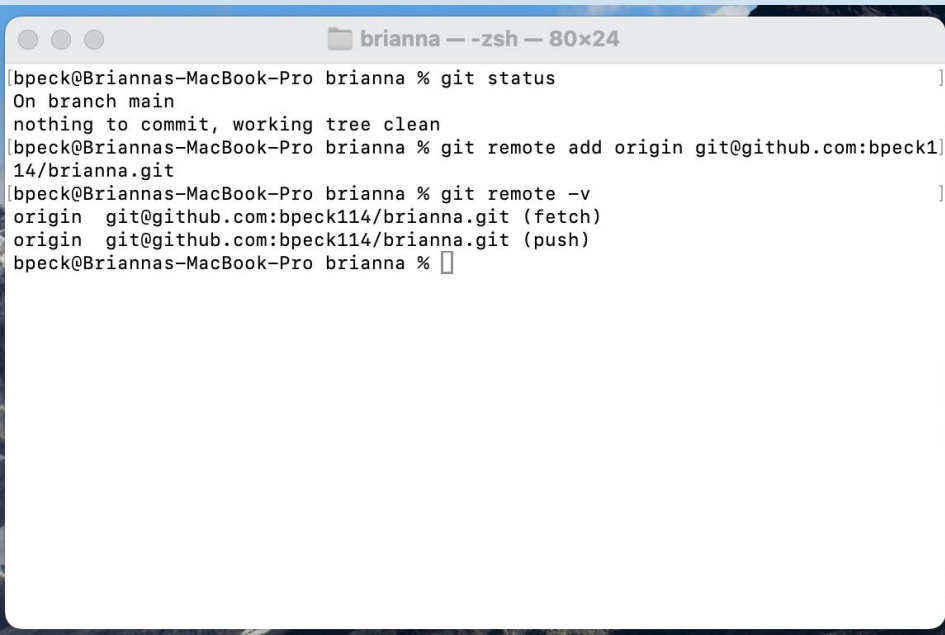
origin = another word for remote repository

Then we see the link we pasted into our command line

fetch = the remote repository where save branches

push = the remote repository where save commits

CHECK THE CONNECTION



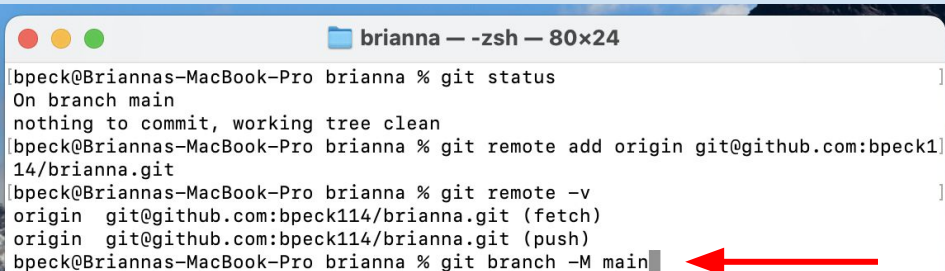
```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck114/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna %
```

Our local and remote repositories are now in communication

But no information has been sent between them

There are a couple more steps

set Remote Branch as main



```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck114/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % git branch -M main
```

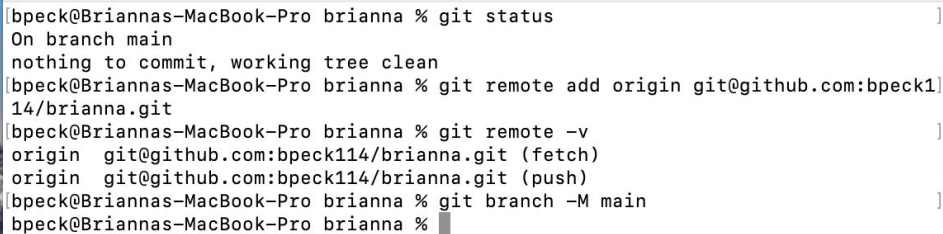
To make sure that we are pushing to the proper branch

Call:
git branch -M main

You don't need to know what branches are yet, just call the command

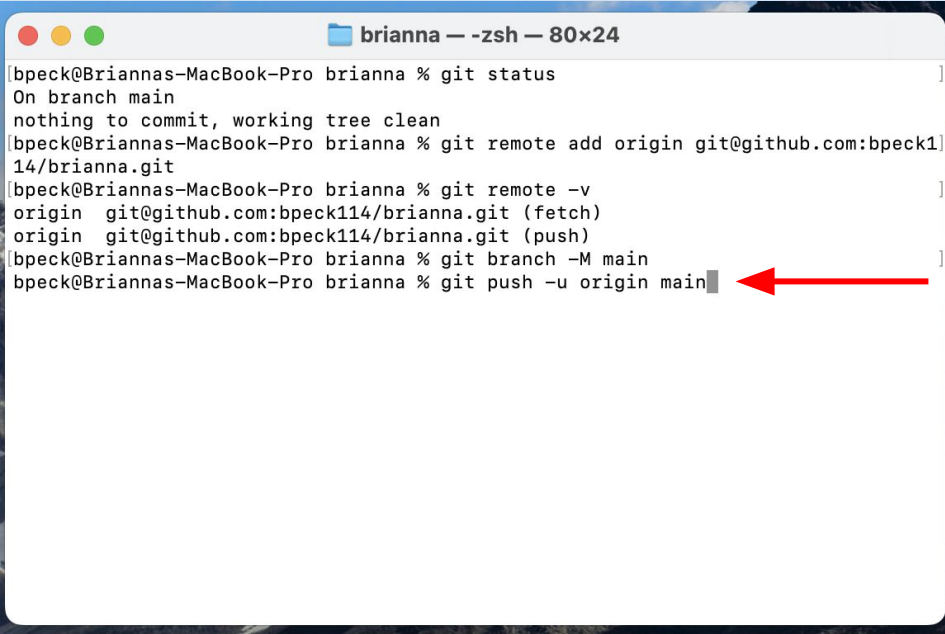
set Remote Branch as main

The last step is to officially save our changes for the first time!



```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck114/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % git branch -M main
bpeck@Briannas-MacBook-Pro brianna %
```

PUSH TO THE Main Branch



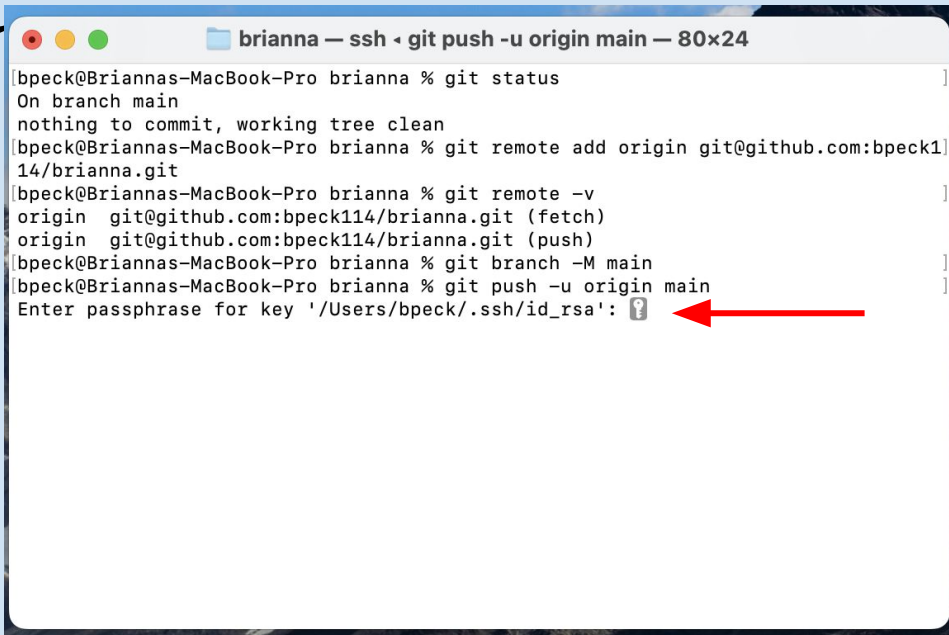
```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck114/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % git branch -M main
bpeck@Briannas-MacBook-Pro brianna % git push -u origin main
```

Type:

git push -u origin main

Now you are officially telling your local repository to save changes to your remote repository

ENTER YOUR PASSWORD



```
brianna — ssh • git push -u origin main — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck1
14/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % git branch -M main
bpeck@Briannas-MacBook-Pro brianna % git push -u origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa': ?
```

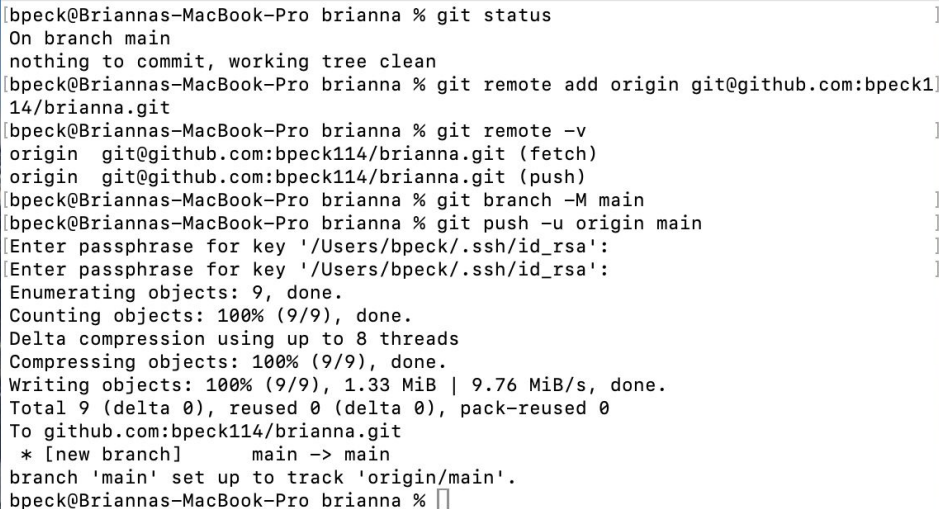
A red arrow points to the question mark in the password prompt.

You will be asked to type in
your SSH password

You created this while
following the Installation
Guide

Nothing will show up for your
privacy!

save your changes Remotely



```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro brianna % git status
On branch main
nothing to commit, working tree clean
bpeck@Briannas-MacBook-Pro brianna % git remote add origin git@github.com:bpeck1
14/brianna.git
bpeck@Briannas-MacBook-Pro brianna % git remote -v
origin  git@github.com:bpeck114/brianna.git (fetch)
origin  git@github.com:bpeck114/brianna.git (push)
bpeck@Briannas-MacBook-Pro brianna % git branch -M main
bpeck@Briannas-MacBook-Pro brianna % git push -u origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 1.33 MiB | 9.76 MiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:bpeck114/brianna.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
bpeck@Briannas-MacBook-Pro brianna %
```

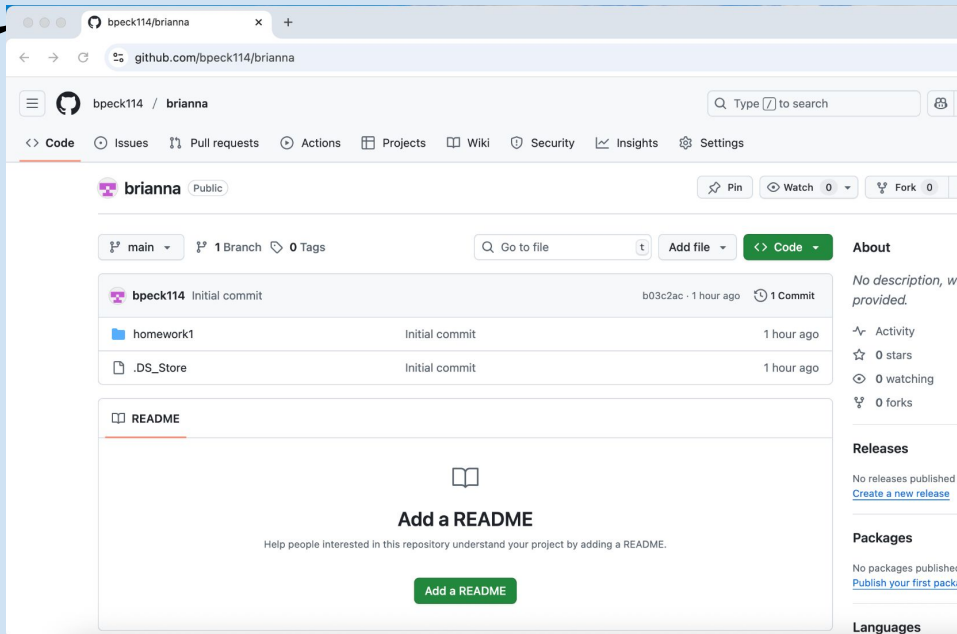
After typing in your password
you will get this really long
message

But as long as your don't see
error in the words, you are
good to go!

save your changes REMOTELY

If you go back to your web browser with GitHub

You should now see your homework1/ folder!





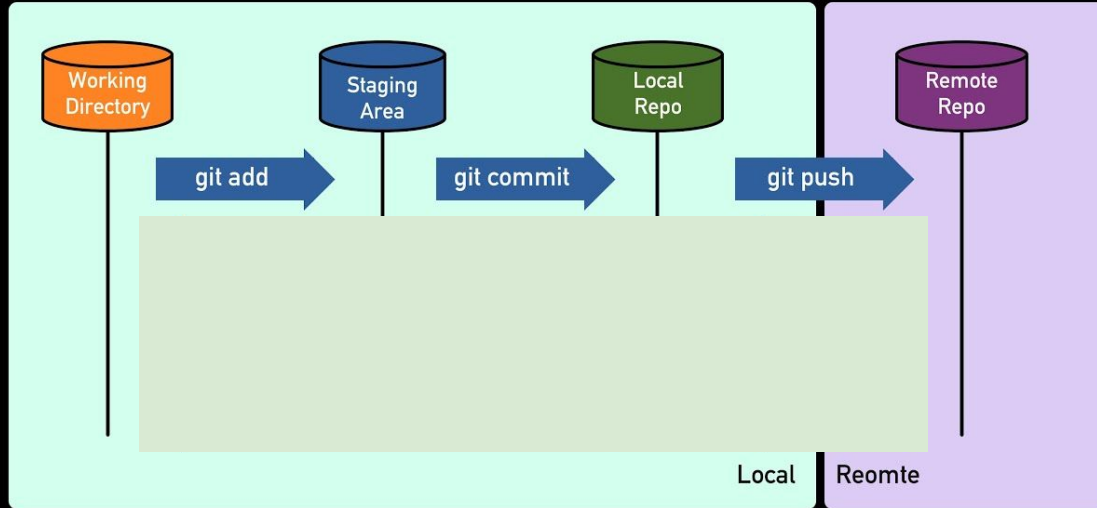
★ 04 ★

Key Takeaways

Most important information

A HELPFUL Diagram

How Git Actually Works



*more of this diagram will be revealed later



BACKGrouND InFormAtION



3 key commands for saving your work with Git/GitHub:

- *git add .* OR *git add <file>*
- *git commit -m "message"*
- *git push origin main*

Follow those three steps anytime you want to save changes!

•





BACKGround InFormAtion



Oh no! I ran into a problem, how do I fix it?

Call these commands to learn more about the issue:

- *git status*
- *git log* OR *git log -1*



USE GITHUB FOR Homework



From now on, you will be turning in your homework via GitHub repositories on Gradescope

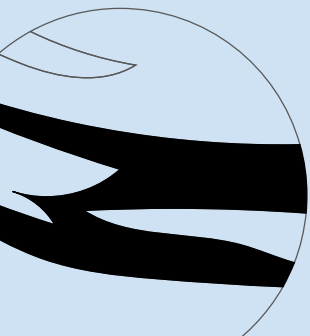
Homework #2 will walk you through how to do so

If you need help, please see the reference slides at the bottom of this slideshow



The background is a light blue gradient. It features several decorative elements: four large, black, four-pointed stars; two smaller, white, four-pointed stars; and three constellations represented by dotted lines connecting small circles. One constellation is in the top left, another in the top right, and a third in the bottom right. A curved dotted line is also visible in the bottom right corner.

Let's Practice





PRACTICE QUESTION



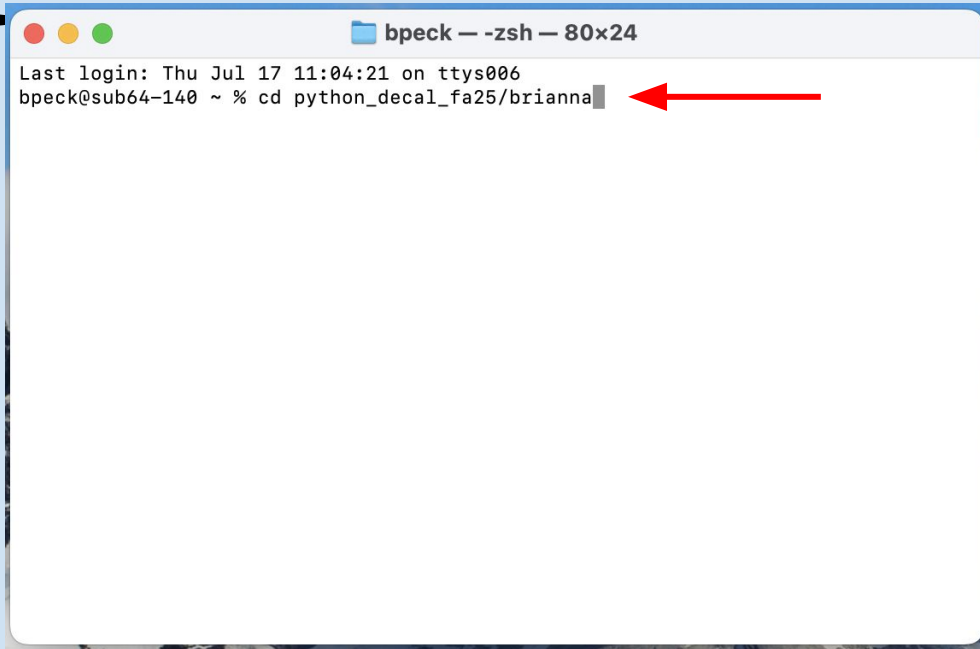
Think by yourself and then discuss these questions with a partner:

- Then we will go through it together as a class

Let's say you just opened the terminal, what commands would you need to call to do the following:

- Enter the yourname directory
 - Create a new file called test.py and
 - Code: `print(5 + 5)`
 - Run the Python code on the terminal
 - Save your changes to GitHub
 - - *Hint:* `git add`, `git commit`, `git push`
- 
- 
- 

Answer: CHANGE DIRECTORIES

A screenshot of a terminal window titled "bpeck — -zsh — 80x24". The window shows the output of a login session: "Last login: Thu Jul 17 11:04:21 on ttys006" followed by the prompt "bpeck@sub64-140 ~ %". The user has entered the command "cd python_decal_fa25/brianna", and a red arrow points to the end of the command line.

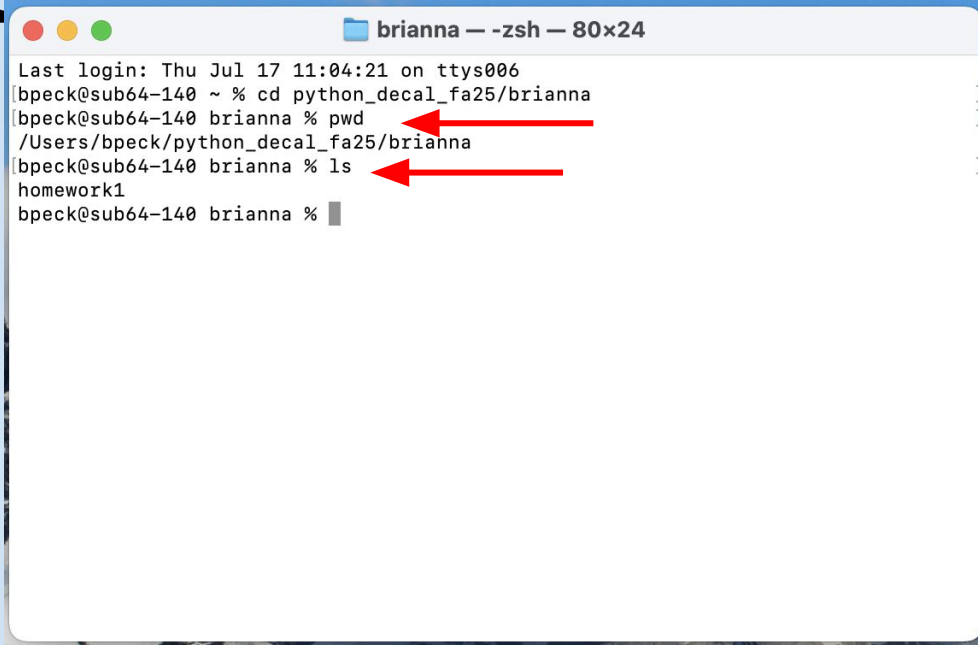
```
bpeck — -zsh — 80x24
Last login: Thu Jul 17 11:04:21 on ttys006
bpeck@sub64-140 ~ % cd python_decal_fa25/brianna
```

Start by opening up the terminal

Then write the file path to your yourname directory

Use the cd command to change directories

Answer: call PWD and LS



A terminal window titled "brianna — -zsh — 80x24" showing a series of commands and their outputs. The user navigates to the directory "python_decad_fa25/brianna" and then runs "pwd" and "ls". Red arrows point to the "pwd" and "ls" commands. The output of "pwd" is "/Users/bpeck/python_decad_fa25/brianna" and the output of "ls" is "homework1".

```
Last login: Thu Jul 17 11:04:21 on ttys006
bpeck@sub64-140 ~ % cd python_decad_fa25/brianna
bpeck@sub64-140 brianna % pwd
/Users/bpeck/python_decad_fa25/brianna
bpeck@sub64-140 brianna % ls
homework1
bpeck@sub64-140 brianna %
```

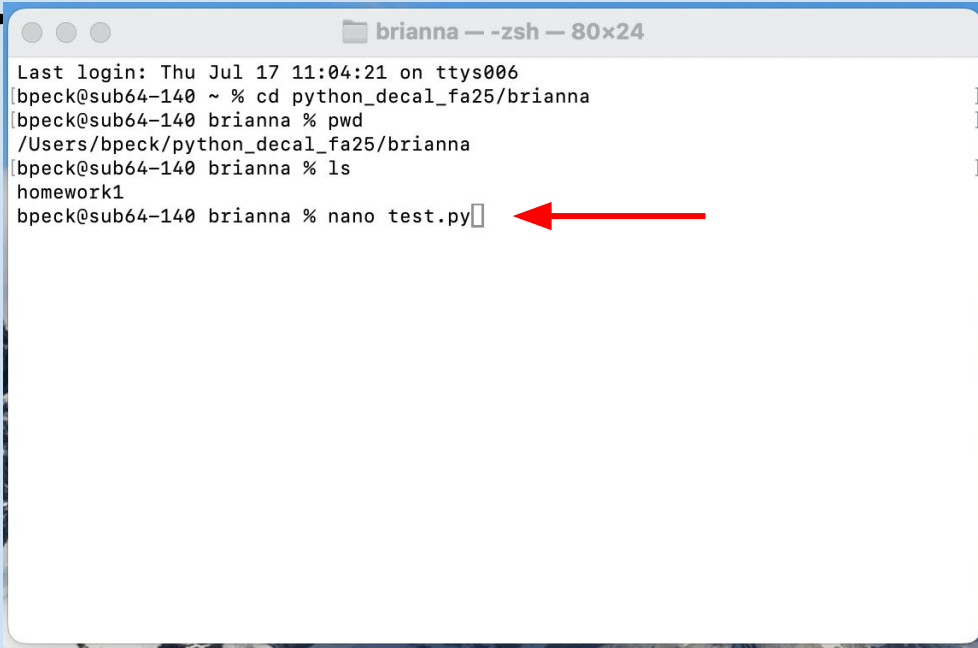
To double check we are in the right location, it is always good to call:

pwd

ls

To figure out your file path and the contents of your directory

Answer: use nano TO edit

A terminal window titled 'brianna -- zsh -- 80x24' showing a series of commands and their outputs. The commands are: 'cd python_decal_fa25/brianna', 'pwd' (output: '/Users/bpeck/python_decal_fa25/brianna'), 'ls' (output: 'homework1'), and 'nano test.py'. A red arrow points to the 'nano test.py' command.

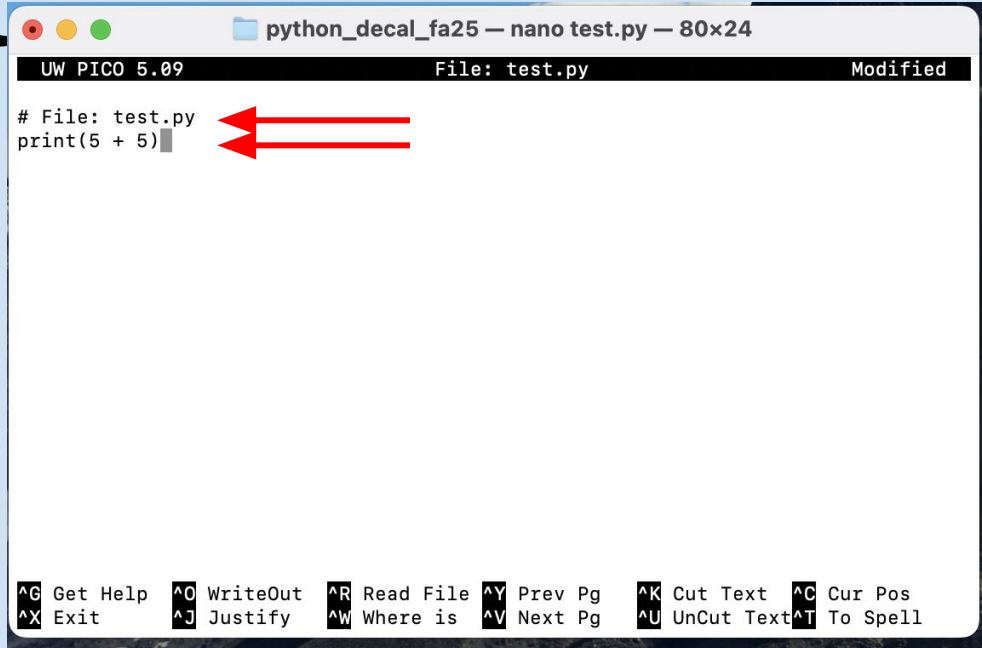
```
brianna -- zsh -- 80x24
Last login: Thu Jul 17 11:04:21 on ttys006
[bpeck@sub64-140 ~ % cd python_decal_fa25/brianna
[bpeck@sub64-140 brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
[bpeck@sub64-140 brianna % ls
homework1
[bpeck@sub64-140 brianna % nano test.py
```

We can immediately open a file by calling:

nano test.py

But you can also use the touch command if you don't want to edit the file

EDIT THE FILE WITH nano



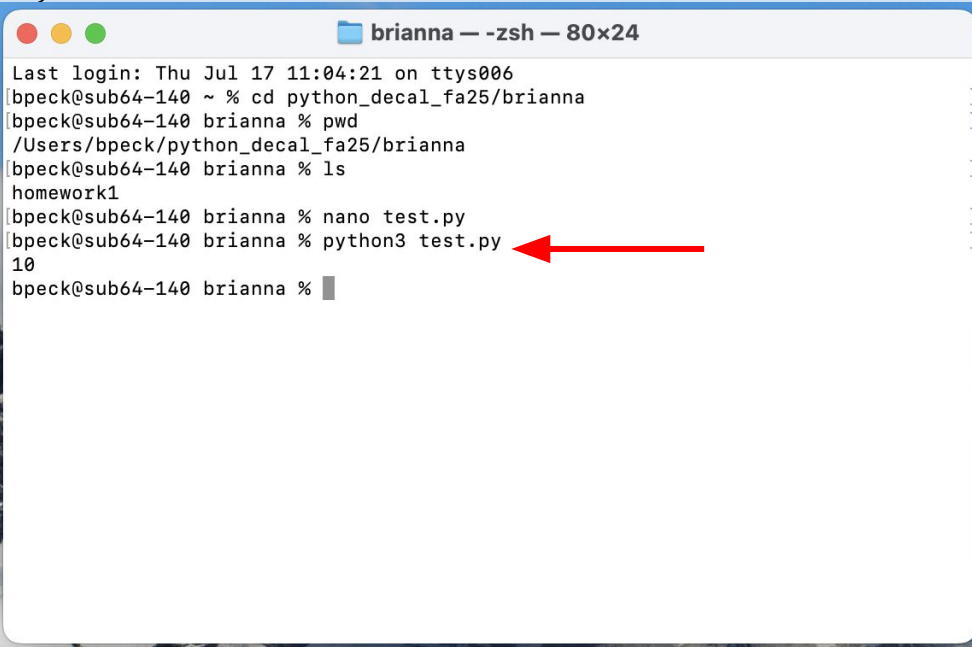
```
python_decal_fa25 — nano test.py — 80x24
UW PICO 5.09      File: test.py      Modified
# File: test.py
print(5 + 5)
```

It's always good to add a comment at the top of the file indicating it's file name

Then type:
print(5 + 5)

Save your changes and close
nano

Answer: RUN SCRIPT IN PYTHON ✨

A screenshot of a terminal window titled 'brianna — -zsh — 80x24'. The window shows a series of commands and their outputs. A red arrow points to the command 'python3 test.py'.

```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:04:21 on ttys006
bpeck@sub64-140 ~ % cd python_decal_fa25/brianna
bpeck@sub64-140 brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@sub64-140 brianna % ls
homework1
bpeck@sub64-140 brianna % nano test.py
bpeck@sub64-140 brianna % python3 test.py
10
bpeck@sub64-140 brianna %
```

To run your script directly on the terminal, you can call:

python3 test.py

Then your terminal will return the operation

Answer: RUN SCRIPT IN PYTHON ✨

brianna — -zsh — 80x24

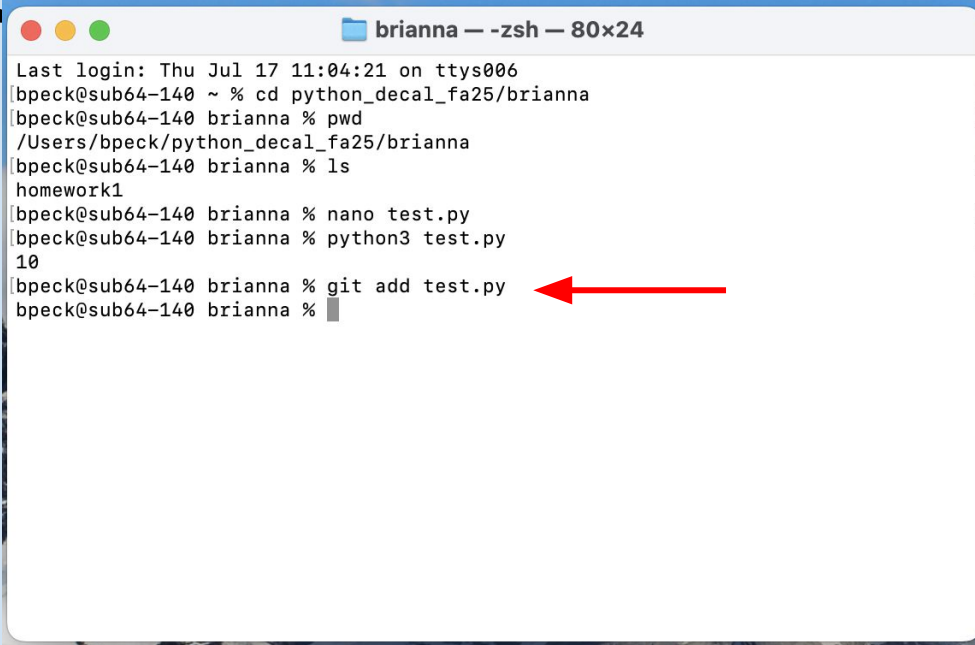
```
Last login: Thu Jul 17 11:04:21 on ttys006
bpeck@sub64-140 ~ % cd python_decal_fa25/brianna
bpeck@sub64-140 brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@sub64-140 brianna % ls
homework1
bpeck@sub64-140 brianna % nano test.py
bpeck@sub64-140 brianna % python3 test.py
10
bpeck@sub64-140 brianna %
```

To save our changes with git,
we need to follow three steps:

- git add
- git commit
- git push

With needed modifications, of
course

Answer: ADD CHANGES Area

A terminal window titled 'brianna — zsh — 80x24' showing a series of commands and their outputs. The commands are: 'cd python_decal_fa25/brianna', 'pwd' (output: '/Users/bpeck/python_decal_fa25/brianna'), 'ls' (output: 'homework1'), 'nano test.py', 'python3 test.py' (output: '10'), and 'git add test.py'. A red arrow points to the 'git add test.py' command. The prompt 'bpeck@sub64-140' is visible at the start of each line.

```
bpeck@sub64-140 ~ % cd python_decal_fa25/brianna
bpeck@sub64-140 brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
bpeck@sub64-140 brianna % ls
homework1
bpeck@sub64-140 brianna % nano test.py
bpeck@sub64-140 brianna % python3 test.py
10
bpeck@sub64-140 brianna % git add test.py
bpeck@sub64-140 brianna %
```

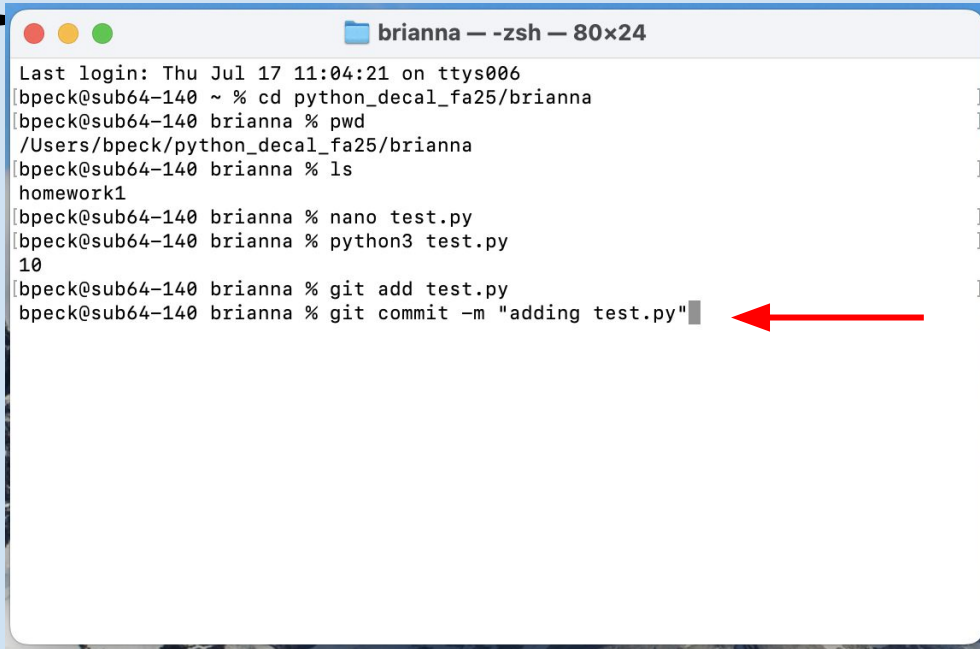
First up is git add

Let's practice only adding the file we created

Type:

git add test.py

Answer: save Locally



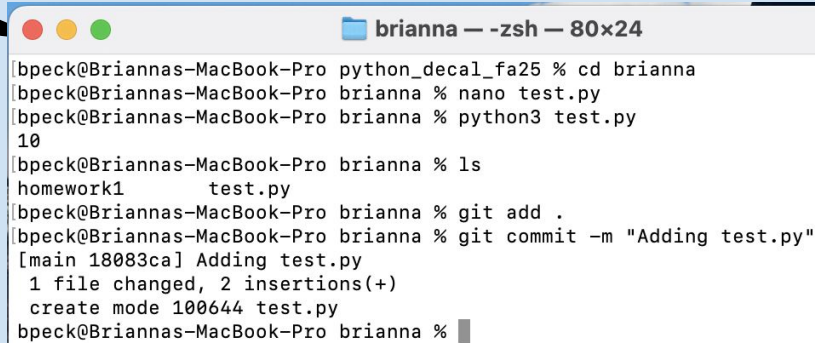
```
brianna — -zsh — 80x24
Last login: Thu Jul 17 11:04:21 on ttys006
[bpeck@sub64-140 ~ % cd python_decal_fa25/brianna
[bpeck@sub64-140 brianna % pwd
/Users/bpeck/python_decal_fa25/brianna
[bpeck@sub64-140 brianna % ls
homework1
[bpeck@sub64-140 brianna % nano test.py
[bpeck@sub64-140 brianna % python3 test.py
10
[bpeck@sub64-140 brianna % git add test.py
[bpeck@sub64-140 brianna % git commit -m "adding test.py" ←
```

To save our changes locally,
we need to use git commit

This moves our changes from
the staging area to version
history

Type:
git commit -m "adding test.py"

Answer: save Locally

A screenshot of a terminal window titled 'brianna — -zsh — 80x24'. The window shows a series of commands and their outputs. The user navigates to the 'brianna' directory, creates a file 'test.py' using 'nano', runs 'python3 test.py' which outputs '10', lists files with 'ls' showing 'homework1' and 'test.py', and finally adds and commits the file using 'git add .' and 'git commit -m "Adding test.py"'. The commit message shows '1 file changed, 2 insertions(+)' and 'create mode 100644 test.py'.

```
[bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
brianna % nano test.py
brianna % python3 test.py
10
brianna % ls
homework1      test.py
brianna % git add .
brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
1 file changed, 2 insertions(+)
create mode 100644 test.py
brianna %
```

The last step is to push or saved history from our local computer to our remote repository

Answer: save Remotely

```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
bpeck@Briannas-MacBook-Pro brianna % nano test.py
bpeck@Briannas-MacBook-Pro brianna % python3 test.py
10
bpeck@Briannas-MacBook-Pro brianna % ls
homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
```

When you call git push, make sure you are pushing to the correct branch

Which for us is main

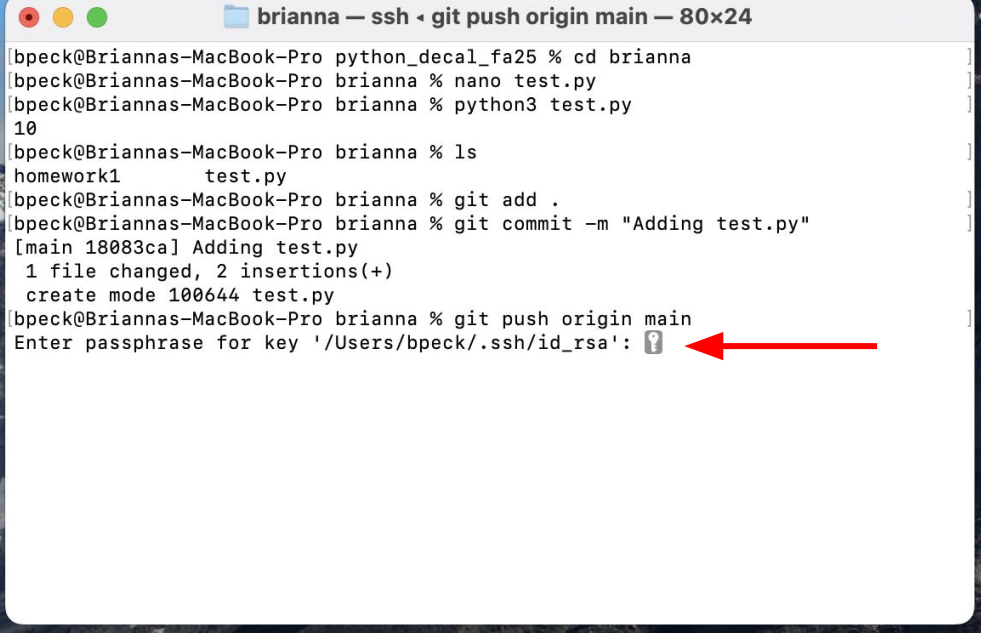
So call:

git push origin main

Answer: Enter Password

You will be asked to type in
your password again

Hidden for your privacy



```
brianna — ssh • git push origin main — 80x24
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
bpeck@Briannas-MacBook-Pro brianna % nano test.py
bpeck@Briannas-MacBook-Pro brianna % python3 test.py
10
bpeck@Briannas-MacBook-Pro brianna % ls
homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
1 file changed, 2 insertions(+)
create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa': ?
```

A red arrow points to the question mark in the password prompt.

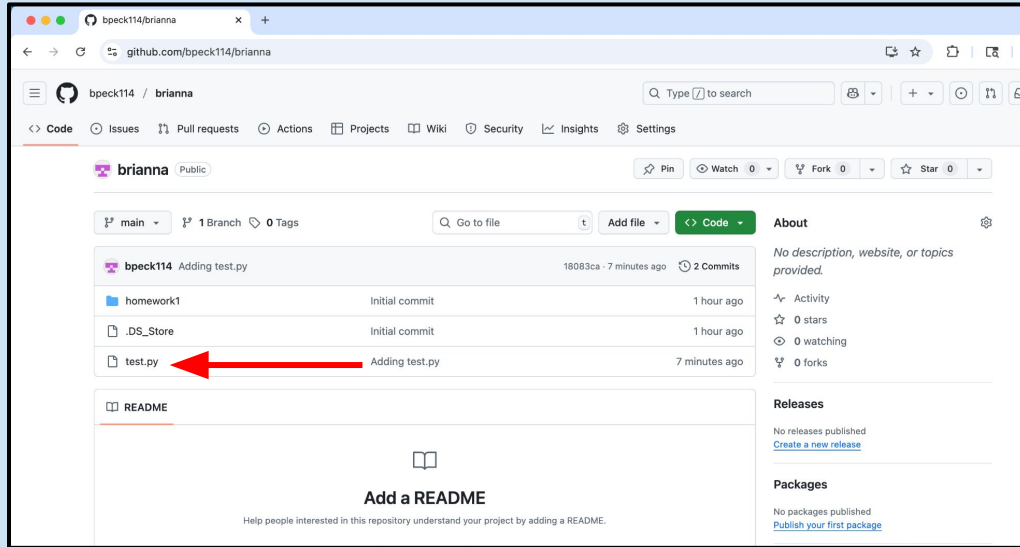
Answer: confirm no errors

```
brianna — -zsh — 80x24
bpeck@Briannas-MacBook-Pro python_decal_fa25 % cd brianna
bpeck@Briannas-MacBook-Pro brianna % nano test.py
bpeck@Briannas-MacBook-Pro brianna % python3 test.py
10
bpeck@Briannas-MacBook-Pro brianna % ls
homework1      test.py
bpeck@Briannas-MacBook-Pro brianna % git add .
bpeck@Briannas-MacBook-Pro brianna % git commit -m "Adding test.py"
[main 18083ca] Adding test.py
 1 file changed, 2 insertions(+)
 create mode 100644 test.py
bpeck@Briannas-MacBook-Pro brianna % git push origin main
Enter passphrase for key '/Users/bpeck/.ssh/id_rsa':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 346 bytes | 346.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:bpeck114/brianna.git
 b03c2ac..18083ca main -> main
bpeck@Briannas-MacBook-Pro brianna %
```

If no error message appears
then you have properly saved
your changes

Answer: CHECK ON GITHUB

When you refresh GitHub, you should now see test.py!

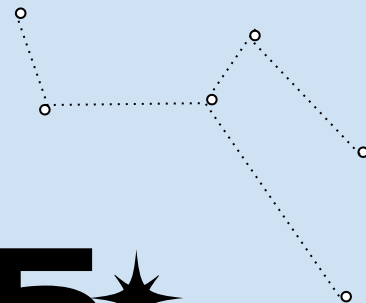




★ 05 ★

Lecture Engagement

Show us what you know!



LECTURE ENGAGEMENT

Fill out Lecture Check 3!

Always due two days after class.

Takes <5 minutes to complete.

